

Capital-Embodied Innovation and Declining Labor Share *

Hyejin Park[†]

Younghun Shim[‡]

June 11, 2026

Abstract

This paper develops an occupation-level measure of Capital-Embodied Innovation (CEI). The effect of CEI on labor-capital substitution is heterogeneous, depending on the similarity between capital and occupational tasks. Specifically, CEI directed toward task-similar capital (CEI-s) triggers a stronger displacement of labor by capital compared to CEI directed toward task-dissimilar capital (CEI-d). Furthermore, occupational exposure to CEI varies in both magnitude and composition. Occupations with higher labor costs relative to capital are subject to higher levels of CEI-s but lower levels of CEI-d. Ultimately, CEI accounts for more pronounced declines in labor share observed in these occupations.

Keywords: Capital-Embodied Innovation, Text Analysis of Patents, Substitution between Labor and Capital

JEL codes: E22, E24, O33, O47

*We thank Ufuk Akcigit, Julieta Caunedo (discussant), Erik Hurst, Greg Kaplan, Keun Lee, Josh Morris-Levenson, Simon Mongey, Robert Shimer, Yongseok Shin, Nancy Stokey, and seminar participants at Southern Economic Association 2023, Korea Development Institute 2023 Workshop, NBER Summer Institute Meeting 2023, NASMES 2023, CESifo 2023, Midwest Macro 2023, Chung-Ang U., Korea U., SNU, Yonsei, SED 2022, Hong Kong Baptist U. for their helpful comments. The paper has been previously circulated under the title “How Task-Biased is Capital-Embodied Innovation?” The views expressed herein are those of the authors and should not be attributed to the IMF, its Executive Board, or its management. The authors gratefully acknowledge the financial support from the Social Sciences and Humanities Research Council in Canada [PV153480, PVX20020]. All errors are ours.

[†]Université de Montréal; Email Address: hyejin.park@umontreal.ca

[‡]International Monetary Fund; Email Address: yshim@imf.org

1 Introduction

In recent decades, a persistent decline in the labor share of income in advanced economies has sparked a central debate about the cause behind this phenomenon (Elsby et al., 2013; Autor et al., 2020). Although a significant body of research has examined technological progress—specifically falling equipment prices and the expansion of automation—as a primary driver of this trend (Karabarbounis and Neiman, 2014; Acemoglu and Restrepo, 2019), how innovation shapes capital-labor substitution under heterogeneous occupational tasks remains under-explored. The recent literature emphasizes that occupations face differential exposure to technological changes (Caunedo et al., 2023; Kogan et al., 2023), but the implications of these interactions for the labor share have not yet been fully elucidated. Identifying these heterogeneous effects is essential for designing R&D policies that reconcile productivity growth with equitable labor market outcomes.

This paper develops an occupation-level measure of Capital-Embodied Innovation (CEI) and examines its heterogeneous effects on labor shares. We classify the heterogeneous relation between capital goods and labor into two categories. A capital good is defined as task-similar if its primary function is similar enough to the core tasks of an occupation. In contrast, it is defined as task-dissimilar if it is utilized within the occupation but performs functions distinct from the core tasks. This classification is implemented by computing text-similarity scores between Wikipedia descriptions of capital goods and O*NET task statements. Then, CEI is quantified by matching patents to the capital goods, based on the text similarity between patent abstracts and their corresponding Wikipedia descriptions. Finally, we incorporate the measure into a structural model of occupation-level production functions to evaluate the aggregate impact of CEI on labor shares.

Our approach differs from existing measures of occupational exposure to innovation. Rather than matching patents directly to task descriptions, as in Kogan et al. (2023) and Autor et al. (2024), we use capital goods as a bridge between patents and occupations. This approach has three advantages. First, it captures innovations that affect an occupation even when the improved capital good does not directly overlap with occupational tasks. Second, it allows capital-embodied changes to heterogeneously affect capital-labor

substitution across occupations. Finally, we can naturally integrate capital data to see if innovation encourages the substitution of labor with capital. These three features are directly related to our main empirical findings.

Our results underscore that innovation often influences occupations even when the improved capital good lacks a direct lexical or semantic connection to specific occupational tasks. For example, although agricultural technicians rely on combine harvesters, their core tasks—such as equipment maintenance and research coordination—do not explicitly mirror the mechanical functions of combine harvesters. A direct patent-to-task matching approach would therefore miss some innovations that are relevant for this occupation. Quantitatively, this task-dissimilar margin was more important than the channel through which innovation is embodied in task-similar capital. Between 1980 and 2015, the majority of technological progress during this period was concentrated in capital goods that are disproportionately task-dissimilar. For instance, information and computing equipment—categories characterized by high rates of innovation—are predominantly task-dissimilar for a wide range of occupations. In contrast, traditional industrial machinery, which is more likely to be task-similar, experienced relatively lower levels of innovation.

Furthermore, the distinction between task-similar and task-dissimilar capital is central to understanding the heterogeneous impacts of innovation on the labor market. Continuing our earlier example, a combine harvester is task-dissimilar for agricultural technicians because its mechanical functions do not directly replicate their core tasks. Innovation in this machinery is therefore more likely to augment their productivity than to displace their labor. By contrast, the same machine is task-similar for farm laborers, whose core activities are directly mirrored by its primary functions, making displacement more likely. Consistent with this intuition, our analysis finds that innovation in task-similar capital (CEI-s) reduces labor demand, whereas innovation in task-dissimilar capital (CEI-d) increases it. These effects operate through two channels. First, task-similar capital is more substitutable with labor: when CEI-s lowers capital user costs, substitution away from labor outweighs the expansion in occupational service production. By

contrast, task-dissimilar capital is more complementary, so the productivity-driven scale effect dominates the substitution effect. Second, CEI-s reduces occupational labor demand even after controlling for changes in capital user costs, consistent with the automation of existing tasks, as in [Acemoglu and Restrepo \(2018a\)](#). CEI-d, by contrast, is associated with the reinstatement effect: the creation of new and complex tasks within the occupation.

Lastly, the data reveal innovation indeed promotes the substitution of capital with labor within occupations. Following the imputation methodology of [Caunedo et al. \(2023\)](#), we leverage fixed asset accounts from the Bureau of Economic Analysis (BEA) to construct occupation-level estimates of capital stocks and prices. This step suggests that occupations have substantial heterogeneity in labor share trends; declines in the labor share are most pronounced in occupations characterized by high labor-to-capital cost ratios. Moreover, these occupations exhibit a distinct innovation profile: they are subject to higher levels of CEI-s but lower levels of CEI-d. This pattern is consistent with technological changes directed to substitute labor with capital in occupations where labor is disproportionately expensive relative to capital. Ultimately, our measure provides a robust explanation for how the specific composition of CEI drives the divergent evolution of occupational labor shares observed in the data.

To derive the quantitative implications of CEI on the labor share, we construct a general equilibrium model in which occupational services are produced by a nested CES function that combines occupational labor with task-similar and task-dissimilar capital. This specification allows for heterogeneous elasticities of substitution between labor and each capital type. In our framework, CEI shifts occupational labor demand in two ways. First, by reducing the effective user costs of capital, CEI incentivizes the substitution of labor with capital. Depending on the relative magnitude of elasticities of substitution, lower costs of capital relative to wages associated with CEI can increase or decrease occupational labor demand. Second, by automating existing tasks or creating new tasks in occupations, CEI directly shifts occupational labor demand. CEI on task-similar capital and that on task-dissimilar capital are also allowed to have different effects on occupa-

tional labor demand.

The first-order condition of occupational labor demand is used to estimate model parameters. Since residual shocks to occupational labor demand could be correlated with CEI, wage, and capital intensity at the occupation level, we devise shift-share instruments from academic publications, capital imports, and immigration trends from Latin American countries. Specifically, the expansion of a scientific field accelerates innovation in related patent classes. Likewise, a rise in capital imports increases capital supply for occupations that utilize those goods intensively, while immigration flows from Latin America expand the labor supply in occupations where these workers possess a comparative advantage. The identification of model parameters relies on the exogenous trends in academic publications, imports, and immigration from Latin America, and predetermined heterogeneity in occupational exposure to these trends. The model is flexible enough to capture heterogeneous effects of CEI on demand for occupational labor and capital. Our estimated elasticities of substitution suggest that lower user costs associated with CEI-s have a stronger effect in reducing labor share than CEI in task-dissimilar capital. In addition, we find that CEI-s decreases occupational labor demand—even after controlling for capital substitution—whereas CEI-d exerts a positive effect on demand.

Finally, we evaluate the quantitative importance of CEI for aggregate and occupational labor shares through a counterfactual analysis by holding patenting activities constant at their 1980 levels. Our results demonstrate that CEI effectively replicates both the observed aggregate trends and the cross-sectional patterns in labor share. First, we find that CEI accounts for nearly the entire decline in the aggregate labor share documented in the data. Specifically, CEI-s drives a more substantial contraction in the aggregate labor share than CEI-d. This difference is rooted in the heterogeneous response of occupational income shares: while CEI-d raises the income share of occupations characterized by higher labor shares, CEI-s reduces it. Second, CEI exerts a disproportionately large impact on the labor share declines within high-labor-cost occupations. These occupations are characterized by higher exposure to CEI-s and a greater intensity of task-similar capital. Consequently, without CEI, the counterfactual user costs of task-similar capital

would have risen more sharply for these specific groups. Given that task-similar capital possesses a higher elasticity of substitution with labor, the technology-driven reduction in user costs leads to the more pronounced labor share reduction observed in high-cost occupations.

Related Literature

This paper investigates the technological origins of the global decline in the labor share. Since the 1980s, this trend has been widely attributed to factors such as offshoring, falling capital costs, the expansion of automation, and the emergence of “superstar” firms (Elsby et al., 2013; Karabarbounis and Neiman, 2014; Acemoglu and Restrepo, 2019; Autor et al., 2020). Although capital-driven explanations typically require an aggregate elasticity of substitution greater than one, empirical literature often produces estimates below unity (Antras, 2004; Chirinko and Mallick, 2017; Oberfield and Raval, 2021). Recent work by Hubmer (2023) identifies an elasticity above one but notes that shifting demand for goods plays a counteracting role. We complement this approach by emphasizing capital-embodied innovation. Exploiting occupation-level variation in labor supply and capital imports, we find an elasticity of substitution that exceeds unity. Furthermore, our results highlight that CEI increases the income share of occupations characterized by high labor shares.

Second, this paper contributes to the extensive literature on the consequences of technological progress on the labor market (e.g., Autor et al., 2006; Goos and Manning, 2007; Lee and Shin, 2017; Bárány and Siegel, 2018; Keller and Utar, 2023). A significant strand of this research—including seminal works by Autor and Dorn (2013), Goos et al. (2014), Michaels et al. (2014), and Acemoglu and Restrepo (2022)—primarily studies the emergence of specific capital goods such as computers and robots. These studies generally find that innovation in these areas has suppressed demand for routine and non-abstract occupations, thereby contributing to job polarization. We broaden this research agenda by constructing an innovation measure for an exhaustive range of capital goods at the occupational level. Crucially, while robots and computers are frequent focal points in the

literature, they represented only 0.7% and 3% of total equipment expenditures in 2019, respectively.¹ By expanding our analytical scope beyond these prominent but narrow categories, we provide a more comprehensive and accurate account of how capital-embodied innovation reshapes the broader occupational landscape.

This paper also relates to the extensive literature that examines capital-skill complementarity (Griliches, 1969; Goldin and Katz, 2008; Hornstein et al., 2005). Most existing frameworks assume that workers from different skill groups exhibit varying elasticities of substitution with capital, with these parameters determining how labor demand for each group responds to capital accumulation (Krusell et al., 2000; Berlingieri et al., 2022; Caunedo et al., 2023). In contrast, our analysis shifts the focus by categorizing capital goods into two distinct functional types, each with its own elasticity of substitution with labor. This parsimonious approach captures the complex heterogeneity in complementarity between capital and labor while relying on only two estimated elasticities.

Lastly, this paper contributes to the expanding literature that utilizes textual analysis of patent data to quantify innovation (Argente et al., 2023; Hémous et al., 2025; Zhestkova, 2021; Bloom et al., 2021; Kelly et al., 2021; Mann and Püttmann, 2023). A predominant methodology within this field involves mapping patents directly to occupational task descriptions to gauge exposure to emerging technologies. For instance, Webb (2019) focuses on patents related to artificial intelligence and robotics, while Kogan et al. (2023) broaden the scope to include a more diverse array of technologies. Similarly, Autor et al. (2024) distinguish between labor-augmenting and labor-saving innovations by matching patents with specific occupational titles and micro-tasks, and Hémous et al. (2025) identify automation-specific patent classes using text-based similarity. We diverge from these task-centric frameworks by leveraging O*NET “Tools Used” data to link patents with the specific capital goods utilized across occupations. This methodology allows our innovation measure to capture technological progress that may lack a direct semantic overlap with occupational tasks but remains integral to the production process embedded in cap-

¹The computer expenditure share is from the BEA Fixed Asset Tables, and the robot share is from the 2019 Annual Capital Expenditure Survey of the Census Bureau. Even when combined with related equipment, such as mainframes and storage devices, computer-related equipment makes up 9.7% of total capital expenditures in 2019.

ital. Our results indicate that these technologies are also important in characterizing the recent decline in the labor share.

The remainder of the paper is organized as follows. Section 2 outlines the empirical framework. Section 3 describes the data used for the analysis and the procedure to construct CEI measures. Section 4 discusses the estimation strategy and the results. Section 5 presents the results of counterfactual exercises. Section 6 concludes.

2 Empirical Framework

2.1 Overview

The economy produces final output by combining the industrial outputs. Within each industry, a representative firm generates outputs by combining a range of occupational services. These services are produced using a nested CES production function that combines labor and two distinct forms of capital: task-similar and task-dissimilar capital. Task-similar and task-dissimilar capital are combined with labor under different substitution elasticities. Capital is supplied competitively as a bundle derived from individual capital goods.

The labor market is characterized by perfect mobility between industries within each occupation. Consequently, wages are determined at the occupation level, and workers are indifferent between industries for a given occupation. On the supply side, individual workers select their occupation by maximizing a utility function that accounts for both the prevailing market wage and idiosyncratic preference shocks. The aggregate labor demand for each occupation is derived by summing the industry-level labor demands for that occupation. Lastly, general equilibrium is defined by the set of occupational wages and the allocation of labor across industry-occupation pairs that clear all occupational labor markets.

2.2 Capital Bundle and the measure of CEI

Competitive capital producers combine individual capital goods into bundles of task-similar (s) and task-dissimilar (d) capital for each industry i and occupation o , denoted by k_{jio} for $j \in \{s, d\}$. These bundles are made through the following aggregator with constant returns to scale.

$$k_{jio} = f(x_{jio1}, \dots, x_{jioN}),$$

where x_{jion} is the quantity of capital good n .²

Under the zero profit condition, the user cost of the capital bundle is defined as:

$$r_{jio} = \sum_{n \in \mathbb{N}_{jo}} \lambda_{jion}^k \frac{x_{jion}}{k_{jio}}, \quad (1)$$

where λ_{jion}^k represents the user cost of capital good n , and $\mathbb{N}_{jo}$ is the set of capital goods that are categorized as group j for occupation o .

Later, we measure CEI using changes in the number of patents matched to specific capital goods, $\# \text{Patent}_{jon}$, which will be detailed in Section 3.3. If patents influence the user costs of capital goods isoelastically—specifically $d \log \lambda_{jion}^k \propto \gamma_P d \log (\# \text{Patent}_{jon})$ —the change in the bundle's user cost can be expressed as:

$$d \log r_{jio} = \underbrace{\gamma_P \sum_n \frac{\lambda_{jion}^k x_{jion}}{r_{jio} k_{jio}} d \log \# \text{Patent}_{jon}}_{=: d \log P_{jio}}. \quad (2)$$

Equation (2) defines the CEI measure at the occupation-level, $d \log P_{jio}$. We approximate $d \log P_{jio}$ using the average change in patent counts between 1980 and 2015 between capital goods, weighted by expenditure shares.

$$d \log P_{jio} \approx \Delta \log P_{jio} \equiv \sum_n \bar{\omega}_{jion} \Delta \log \# \text{Patent}_{jon}. \quad (3)$$

²In the counterfactual analysis, we assume $f(\cdot)$ is a CES aggregator with an elasticity of substitution with $\phi = 1.13$, following [Caunedo et al. \(2023\)](#).

In this equation, $\bar{\omega}_{jon}$ is the average expenditure share of the capital good n between 1980 and 2015, and $\Delta \log \# \text{Patent}_{jon}$ is the logarithmic difference in the number of patents associated with capital goods. A negative γ_p implies that the user costs of capital decrease with CEI. It is important to note that the coefficient γ_p does not vary across task-similar and task-dissimilar capital groups. From now on, we refer to $\Delta \log P_{sio}$ as CEI-s and $\Delta \log P_{dio}$ as CEI-d.

2.3 Occupational Production Function

The aggregate output Y is a Cobb-Douglas composite of industrial outputs, Y_i .

$$Y = \prod_i (Y_i)^{\alpha_i}.$$

Industrial output in industry i , Y_i , aggregates occupational services with a constant elasticity of substitution, σ .

$$Y_i = \left(\sum_o \mu_{io}^{\frac{1}{\sigma}} y_{io}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (4)$$

where μ_{io} is the shifter for occupational labor demand for industry i and occupation o . Occupational service y_{io} is produced with capital and labor as in the following equations.

$$\Theta_{io} = \left(z_{dio}^{\frac{1}{\rho_d}} k_{dio}^{\frac{\rho_d-1}{\rho_d}} + l_{io}^{\frac{\rho_d-1}{\rho_d}} \right)^{\frac{\rho_d}{\rho_d-1}}, \quad (5)$$

$$y_{io} = \left(z_{sio}^{\frac{1}{\rho_s}} k_{sio}^{\frac{\rho_s-1}{\rho_s}} + \Theta_{io}^{\frac{\rho_s-1}{\rho_s}} \right)^{\frac{\rho_s}{\rho_s-1}}. \quad (6)$$

In these equations, k_{dio} and k_{sio} represent task-dissimilar and task-similar capital with their respective levels of productivity, z_{dio} and z_{sio} . l_{io} denotes labor inputs. This nested CES structure follows [Krusell et al. \(2000\)](#), allowing for different substitutability between various production inputs. ρ_d and ρ_s are the substitution elasticity of labor with task-dissimilar and task-similar capital, respectively. This structure implies that the substi-

tution elasticity between the task-dissimilar capital and the task-similar capital is also ρ_s .

A representative firm of industry i chooses labor and capital inputs to minimize production costs, given the user costs of task-similar and task-dissimilar capital, r_{sio} and r_{dio} , and the occupational wage w_o . The first-order conditions govern the relative input intensity as in the following equations.

$$\frac{r_{dio}}{w_o} = z_{dio}^{\frac{1}{\rho_d}} \left(\frac{k_{dio}}{l_{io}} \right)^{-\frac{1}{\rho_d}}, \quad (7)$$

$$\frac{r_{sio}}{w_o} = \Theta_{io}^{\frac{\rho_d - \rho_s}{\rho_s \rho_d}} z_{sio}^{\frac{1}{\rho_s}} k_{sio}^{-\frac{1}{\rho_s}} l_{io}^{\frac{1}{\rho_d}}, \quad (8)$$

$$\frac{w_o}{w_{o'}} = \left(\frac{\mu_{io}}{\mu_{io'}} \right)^{\frac{1}{\sigma}} \left(\frac{y_{io}}{y_{io'}} \right)^{-\frac{1}{\sigma} + \frac{1}{\rho_s}} \left(\frac{\Theta_{io}}{\Theta_{io'}} \right)^{\frac{\rho_s - \rho_d}{\rho_s \rho_d}} \left(\frac{l_{io}}{l_{io'}} \right)^{-\frac{1}{\rho_d}}. \quad (9)$$

Combining these three equations gives the following equation that governs the relative demand for labor within industry i .

$$\frac{w_o}{w_{o'}} = \left(\frac{\mu_{io}}{\mu_{io'}} \right)^{\frac{1}{\sigma}} \left(\frac{\tilde{y}_{io}}{\tilde{y}_{io'}} \right)^{-\frac{1}{\sigma} + \frac{1}{\rho_s}} \frac{\tilde{\Theta}_{io}^{\frac{\rho_s - \rho_d}{\rho_s \rho_d}}}{\tilde{\Theta}_{io'}^{\frac{\rho_s - \rho_d}{\rho_s \rho_d}}} \left(\frac{l_{io}}{l_{io'}} \right)^{-\frac{1}{\sigma}}. \quad (10)$$

In these equations, $\tilde{\Theta}_{io} = \Theta_{io}/l_{io}$ and $\tilde{y}_{io} = y_{io}/l_{io}$ denote the efficiency of labor for the inner and outer composites of occupational service production. In equilibrium, they can be expressed as follows.

$$\tilde{\Theta}_{io} = \left(z_{dio} \left(\frac{r_{dio}}{w_o} \right)^{1 - \rho_d} + 1 \right)^{\frac{\rho_d}{\rho_d - 1}}, \quad (11)$$

$$\tilde{y}_{io} = \tilde{\Theta}_{io}^{\frac{\rho_d - \rho_s}{\rho_d}} \left(z_{sio} \left(\frac{r_{sio}}{w_o} \right)^{1 - \rho_s} + \tilde{\Theta}_{io}^{\frac{\rho_s - 1}{\rho_d}} \right)^{\frac{\rho_s}{\rho_s - 1}}. \quad (12)$$

$\tilde{\Theta}_{io}$ and $\tilde{y}_{io}$ decrease unambiguously with r_{sio} and r_{dio} , respectively. In other words, as capital costs fall, labor efficiency increases for the inner and outer composites of occupational service production.

Equation (10) shows that the relative magnitudes of the substitution elasticities shape how capital-embedded changes affect the demand for labor between occupations. As in [Caunedo et al. \(2023\)](#), a decrease in user costs of task-similar capital leads to substitution of labor with task-similar capital. At the same time, it increases $\tilde{y}_{io}$ and the demand for occupational service o through lower costs of occupational service production. If $\rho_s > \sigma$, the labor substitution induced by a lower task-similar capital cost outweighs the scale effect of increased service demand, thus reducing the labor demand for occupation o relative to others. In contrast, if $\sigma > \rho_d$, the overall increase in the demand for occupational services can be stronger than the substitution with task-dissimilar capital. An increase in $\tilde{\Theta}_{io}$ from lower user costs of task-similar capital raises both $\tilde{y}_{io}$ and $\tilde{\Theta}_{io}$. If $d \log \tilde{y}_{io} / d \log \tilde{\Theta}_{io}$ is close to one or if the occupation does not have task-similar capital, $\sigma > \rho_d$ implies that lower user costs of task-dissimilar capital raise relative labor demand. Thus, in this framework, the effect of CEI on labor demand through changes in capital costs depends on the relative magnitudes of elasticities between capital and labor, as well as across occupational services.

We also allow CEI to directly affect the demand for occupational labor. We assume that the same CEI measure in Equation (2), $d \log P_{jio}$, changes the demand shifter, μ_{io} :

$$d \log \mu_{io} = \gamma_s d \log P_{sio} + \gamma_d d \log P_{dio}. \quad (13)$$

A positive γ_j implies that the demand for occupational labor increases with CEI- j ($j \in \{s, d\}$), even after accounting for its effect on capital substitution in $\tilde{\Theta}_{io}$ and $\tilde{y}_{io}$. Appendix C.1 provides a microfoundation for this assumption using a task-based model of occupational-service production in which CEI can automate existing tasks or create new ones.

2.4 Varying Elasticity of Substitution with Capital

The elasticity of substitution between capital and labor depends on capital composition, since task-similar and task-dissimilar capital have different substitution elasticities

with labor. If task-similar capital is more substitutable with labor, the capital-labor ratio responds more strongly to changes in the user cost of capital for occupations with a higher intensity of task-similar capital. This implies an endogenous heterogeneity in the elasticity of substitution between capital and labor at the occupation level. To formalize this idea, we first define the occupation-level elasticity of substitution.

$$\hat{\rho}_{io} := -\frac{d \log(k_{io}/l_{io})}{d \log(r_{sio}/w_o)} - \frac{d \log(k_{io}/l_{io})}{d \log(r_{dio}/w_o)}. \quad (14)$$

In this equation, $k_{io} = g(k_{sio}, k_{dio})$ is a constant-return-to-scale aggregator of capital types.³ This elasticity, which reflects the response to a uniform reduction in capital costs relative to wages, can be defined independently of the structural model as in [Oberfield and Raval \(2021\)](#). In Appendix C.3, we show that $\hat{\rho}_{io}$ can be expressed as a weighted average of the two structural elasticities ρ_s and ρ_d plus an adjustment term.

$$\hat{\rho}_{io} := -\rho_s (1 - \mathbf{w}_{dio}) - \rho_d \mathbf{w}_{dio} - (\rho_d - \rho_s) (1 - \mathbf{w}_{dio}) \tilde{\mathbf{w}}_{dio} \quad (15)$$

In this equation, $\mathbf{w}_{dio} = r_{dio}k_{dio}/(r_{dio}k_{dio} + r_{sio}k_{sio})$ is the expenditure share of task-dissimilar capital in total capital expenditure, and $\tilde{\mathbf{w}}_{dio} = r_{dio}k_{dio}/(r_{dio}k_{dio} + w_o l_{io})$ is the expenditure share of task-dissimilar capital in the production of the inner composite. The equation implies that the elasticity at the occupation-level is closer to ρ_s if the share of task-similar capital increases and vice versa. The adjustment term arises because the change in r_{dio}/w_o has the additional effect of substituting inner composite with task-similar capital, thus altering the ratio between labor and task-similar capital. In the limit case where $\tilde{\mathbf{w}}_{dio}$ is close to one, $\hat{\rho}_{io} = \rho_d$ because the shadow price of the inner composite relative to task-similar capital does not change with the proportional change in user costs of capital, and the relevant substitution occurs only between task-dissimilar capital and labor.

Finally, this occupation-level elasticity is closely related to changes in labor share in

³In most papers, including [Krusell et al. \(2000\)](#), $g(\cdot, \cdot)$ is a simple summation; In [Caunedo et al. \(2023\)](#), $g(\cdot, \cdot)$ is a CES aggregator.

response to changes in the ratio between relative costs (Oberfield and Raval, 2021).

$$\hat{\rho}_{io} - 1 = \frac{1}{\text{labor share}_{io}(1 - \text{labor share}_{io})} \frac{d\text{labor share}_{io}}{d(r_{io}/dw_o)} \quad (16)$$

Consequently, the labor share responds more elastically to a change in relative capital costs if the elasticity above unity becomes higher. In Section 5, we show that occupations with higher wages relative to capital costs had a larger share of task-similar capital relative to task-dissimilar capital. Thus, they have a higher occupational elasticity of substitution and a more elastic response to a change in user costs of capital.

3 Data and Measurement

3.1 Data

We construct our CEI measure by merging occupation-capital pairs from O*NET with other data from Wikipedia and the United States Patent and Trademark Office (USPTO). The O*NET “Tools Used” data serve as our primary reference for identifying capital goods within each occupation. This source provides a comprehensive set of the machinery and equipment vital to 775 distinct occupations (Dierdorff et al., 2006).⁴ We use Wikipedia articles to construct functional descriptions for each capital good and information on patents filed at the USPTO to measure innovation. The USPTO data contain approximately 5.7 million patents assigned between 1970 and 2015. Our analysis focuses exclusively on utility patents and excludes design and plant patents to capture only quality-improving innovations. Lastly, we use the O*NET “Task Statements” to compare occupational tasks with capital functions.

Occupation-level employment and wages are sourced from the Decennial Census, the American Community Survey, and the Annual Social and Economic Supplement (CPS-

⁴This study uses version 25.0, which was updated in August 2020, whereas Caunedo et al. (2023) use the Dictionary of Occupational Titles and different versions of “Tools Used” to capture changes in capital goods at the occupation level. We use the version of the “Tools Used” data to be consistent with the Wikipedia data, which is also from 2020. IV estimation addresses the measurement errors associated with changes in capital goods over time.

ASEC) of the Current Population Survey. To account for heterogeneous labor productivity arising from observable worker characteristics, we residualize wages using a Mincerian regression that controls for age, education, race, and year effects. This regression is estimated on a sample of prime-age male workers to minimize selection bias.

For capital stocks and user costs, we use fixed- and current-cost capital estimates from the BEA. These estimates are measured in 2012 prices and provided at the level of industry and the NIPA capital category. Fixed-cost capital serves as the quantity of capital goods, and the ratio between current- and fixed-cost estimates is used to measure the price of capital goods in the calculation of user costs. As BEA data are available at the industry level, we assign them to specific occupations using an imputation method described in Section 3.2.

Finally, we integrate several additional datasets to construct instrumental variables. The patent citation data from Marx and Fuegi (2020) are combined with the academic publication records of the Microsoft Academic Graph (MAG). For capital imports, we rely on the UN Comtrade data to track commodity-level import volumes from 1980 to 2015. Detailed descriptions of the data are provided in Appendix A.1.

3.2 Identifying Task-Dissimilar versus Task-Similar Goods

For each occupation, we categorize capital goods into two categories: task-similar and task-dissimilar capital. We define a capital good as task-similar if its primary functions closely align with the specific tasks of an occupation. Conversely, capital goods utilized within an occupation that do not mirror its core tasks are labeled as task-dissimilar. This classification allows for the possibility that the same capital good may be task-similar for one occupation but task-dissimilar for another. For example, both farm workers and agricultural engineers use combine harvesters, but only farm workers have tasks that can be implemented by combine harvesters. At this point, we only allow the two types of capital to have different elasticities of substitution with labor but do not presuppose their relationships with occupational labor demand before the estimation.

This definition is motivated by the existing literature that matches occupations with patents based on text similarity (e.g., [Webb, 2019](#); [Kogan et al., 2023](#)). These studies often find a strong labor-displacement effect of innovations. However, when new technologies perform functions that differ from occupational tasks but are nonetheless used by those occupations, their effects may differ. Using the O*NET list of capital goods as an intermediary, we identify the effect of new technologies applied to a tool when it performs functions distinct from occupational tasks.

To operationalize this classification, we exploit the degree of textual similarity between O*NET task descriptions and Wikipedia articles on capital goods. We first preprocess the raw text using standard natural language processing techniques, including stopword removal and lemmatization, to ensure that matches are based on substantive content rather than functional words.⁵ We then measure the similarity between each task and each capital good using the cosine similarity of term frequency-inverse document frequency (TF-IDF) vectors. For each word i in document j , the TF-IDF weight is defined as

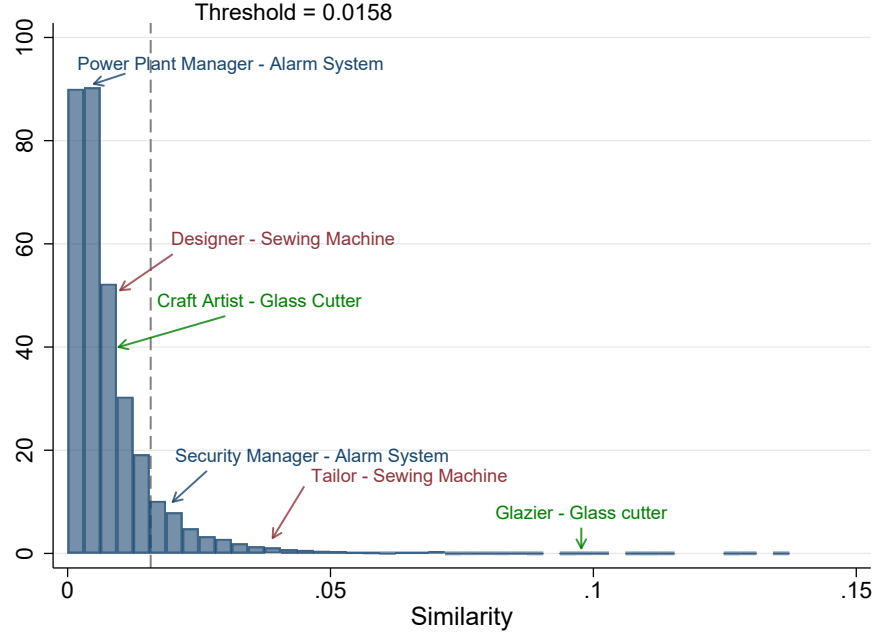
$$w_{ij} = \text{TF}_{ij} \times \text{IDF}_i, \quad \text{TF}_{ij} = \frac{f_{ij}}{\sum_i f_{ij}}, \quad \text{IDF}_i = \log \left(\frac{J}{\sum_j \mathbb{1}\{i \in j\}} \right),$$

where f_{ij} denotes the number of times word i appears in document j , and J is the total number of documents in the corpus. Term frequency (TF_{ij}) measures how often a word i appears in a document j , while inverse document frequency (IDF_i) downweights words i that are common across documents. We then compute cosine similarity between the TF-IDF vectors of a task and a capital good. This measure ranges from 0 to 1, with higher values indicating greater textual similarity. After constructing these scores at the task-capital level, we calculate an unweighted average across all tasks within an occupation to derive a final capital-occupation similarity score.

Figure 1 shows the distribution of the similarity scores between capital goods and occupations. The distribution is right-skewed, indicating that most capital-occupation pairs do not have many overlapping words. However, some capital goods have descriptions closely related to the task descriptions of occupations. For example, in Figure 1, the glass

⁵Appendix B.2 provides details of the cleaning procedure.

Figure 1: Distribution of Similarity Scores



Notes. This figure plots the density of similarity scores for the pairs of capital goods and occupations. The text similarity score is initially measured at the task level for each capital good and then aggregated to the capital-occupation level.

cutter has one of the highest similarity scores with glaziers but a low similarity score with craft artists. Based on these scores, a capital good is considered task-similar to the occupation if the similarity exceeds the 90th percentile; all other capital goods are classified as task-dissimilar.⁶

Finally, we construct occupation-level capital stocks for each category by allocating industry-level BEA capital to occupations. Following [Caunedo et al. \(2023\)](#), this allocation is based on employment shares weighted by capital-use intensity, which we define as the number of specific tools assigned to each NIPA category within an occupation. We then construct a measure of the occupation-level capital bundle as a chained index, starting from the base year 1980 and evolving according to the expenditure-weighted average growth rates across NIPA capital categories. The capital stock is measured as its value in 1980, and the corresponding user costs of the capital bundles are computed by aggregating category-level user costs as in the zero-profit condition of Equation (1). Additional

⁶We try different thresholds and show the estimation results in Appendix D.

details are provided in Appendix A.2.

Table 1: Similar Capital Share

	1980		2015	
	capital	worker	capital	worker
Computers	0.02	0.14	0.04	0.16
Communication eq.	0.23	0.18	0.12	0.20
Medical eq.	0.19	0.35	0.18	0.43
Photocopy eq.	0.07	0.08	0.09	0.07
Office eq.	0.08	0.28	0.12	0.27
Metal products	0.08	0.27	0.08	0.23
Engines	0.27	0.35	0.25	0.31
Aircraft	0.33	0.02	0.23	0.02
Electrical eq.	0.05	0.08	0.10	0.13
Autos	0.33	0.19	0.25	0.19
Ag. machinery	0.33	0.86	0.33	0.73
Serv. machinery	0.22	0.49	0.19	0.43

Notes. This table shows the share of task-similar capital in total expenditure on capital and the fraction of workers with at least one task-similar capital good, conditional on using the capital category.

Table 1 summarizes the share of task-similar capital in total capital expenditure and the percentage of workers using at least one task-similar capital good within those NIPA categories. In general, categories such as engines, heavy equipment, automobiles, and machinery are more likely to be classified as task-similar capital, while computers and electrical equipment typically function as task-dissimilar capital. Capital goods such as aircraft, photocopying machines, and electrical equipment perform functions semantically different from the primary tasks of the workers operating them. In contrast, most workers using agricultural machinery perform tasks that partially overlap with the functions of the machinery itself. Similarly, more than 40% of workers who operate medical equipment and service machinery perform tasks that are partially executed by the corresponding capital goods.

Validation of Occupation-Level Capital Stock Imputation

We perform two exercises to verify the reliability of the capital data imputed at the occupation level.⁷ First, we compute occupational labor shares, defined as the labor expenditure divided by the sum of labor and capital expenditures. Using these numbers, we calculate state-level labor shares as the expenditure-weighted average of occupational labor shares within each state. We then compare these values against state-level labor share data from the BEA. Despite the limitations of occupational labor shares—the exclusion of corporate profits and expenditures on structures—the two measures exhibit a robust correlation coefficient of 0.56 over the 1980–2015 period. Second, we correlate the occupation-level importance of computers from O*NET with the corresponding expenditure shares imputed at the occupation level. This comparison yields a correlation of 0.48. Detailed scatter plots for both validation exercises are presented in Appendix Figure A1.

3.3 Measuring Capital-Embodied Innovation

We quantify Capital-Embodied Innovation (CEI) by matching patents directly to capital goods using a text-based similarity, as in the procedure detailed in Section 3.2. A patent is assigned to a capital good in an eight-digit United Nations Standard Products and Services Code (UNSPSC) if the similarity score of its title and abstract to the Wikipedia description of the capital good exceeds the 90th percentile across patent-capital pairs. Because a patent may be relevant to multiple capital goods or none at all, we also allow patents to be matched to multiple or none of the capital goods based on the similarity score. For patents linked to multiple goods, we restrict each patent to a maximum of five capital goods and apply a weight equal to the inverse of the number of matched goods. After this procedure, 27% of patents are matched with at least one capital good.⁸ We count

⁷Caunedo et al. (2023) establishes the validity of their imputation method through two exercises. First, they utilize the industry shares for each occupation to reconstruct industry-level capital stocks, demonstrating that these estimates are highly correlated with observed BEA records. Second, they illustrate that the proportion of workers using computers—measured at the one-digit occupation level—aligns closely with survey results from the CPS.

⁸Table A4 shows the share of patents that are matched at least one capital good across patent classes and periods, and Example 1 in Appendix B.3 displays a sample matching between patent and capital goods.

Table 2: Summary Statistics of Patents Matched with Capital Goods

	Similar		Dissimilar		N
	Mean	SD.	Mean	SD.	
1970 – 1980	43.1	91.5	44.7	69.9	15,902
1980 – 1990	90.3	174.3	93.5	130.1	15,902
1990 – 2000	330.1	606.2	765.8	1135.8	15,902
2000 – 2015	1052.2	1969.0	2444.6	3275.9	15,902

Notes. This table displays the summary statistics of patents matched with task-dissimilar and task-similar capital aggregated at the occupation-industry level. We take the average number of patents, weighted by the share of capital expenditure in each period.

the cumulative number of patents associated with each capital good from 1970.

Then, the patent counts of each capital good are aggregated to the level of occupation, industry, and capital type. To achieve this, we first compute the unweighted average number of patents within each NIPA capital category for each occupation and capital type, using the crosswalk between UNSPSC and NIPA categories from [Caunedo et al. \(2023\)](#). Occupation-level CEI is then defined as the average log-change in patent counts between 1980 and 2015 across NIPA categories, where each category is weighted by the average capital expenditure shares over the same period, as specified in Equation (3).⁹ By taking averages across capital goods and then capital categories, our CEI measures do not mechanically increase with the variety of capital goods within NIPA categories and the quantity of capital for each NIPA category.

Table 2 presents the summary statistics for the average number of patents associated with occupation, industry, and capital type.¹⁰ The number of patents has increased over time but at different rates between capital types. Initially, the number of patents was comparable between task-similar and task-dissimilar capital. However, task-dissimilar capital experienced faster growth in patents than task-similar capital, suggesting that more

⁹In our context, taking the log difference in the number of patents removes time-invariant components of measurement errors associated with the text-matching procedure. For example, if Wikipedia articles about lasers are easier to match than those about computers and the errors are multiplicatively separable and constant over time, log-differencing the number of patents cancels out the errors.

¹⁰If an occupation does not have any task-similar good, the number of patents is zero for task-similar capital.

patents are made on capital goods used as task-dissimilar capital, such as computers and electrical equipment.

Because pages of certain tools do not exist on Wikipedia and are consequently excluded from our analysis, the CEI for capital goods with higher missing rates may be underestimated. There is substantial variation in these missing rates across the NIPA categories; for example, electrical transmission exhibits the lowest missing rate at 27%, while the aircraft has the highest at 86%. To mitigate this potential bias, we reweight the tools by the inverse of their Wikipedia match rate. We find that the results remain quantitatively similar under this alternative specification, as reported in Appendix Table A4.

4 Estimation

4.1 Strategy

Estimation of γ_p , the coefficient of CEI on capital user costs, is straightforward. We apply Equation (2) to estimate γ_p .

$$\Delta \log r_{jio} = \varsigma_i + \gamma_p \Delta \log P_{jio} + \omega_{jio}. \quad (17)$$

Unless otherwise noted, Δ denotes an operator that takes the difference between 1980 and 2015. r_{jio} represents the user cost of capital, ς_i is the industry fixed effects, and $\Delta \log P_{jio}$ is our measure of CEI defined in Equation 3.¹¹

Then, we use the first-order conditions of cost minimization in Section 2.3 for the estimation of all other parameters. Specifically, Equation (10) allows us to estimate the substitution elasticities for inner and outer CES composites, as well as the elasticity between occupational services. Combining Equations (7) and (8) with (10) yields our empirical

¹¹We estimate the regression at the occupation-industry level, as this intersection represents the main unit of analysis for our empirical study. To ensure the robustness of our results, we alternatively perform the estimation at the NIPA category level and confirm that the resulting coefficients are quantitatively similar to our baseline findings (see Section 4.2).

specification.

$$\begin{aligned}
\Delta \log l_{io} = & \gamma_s \Delta \log P_{sio} + \gamma_d \Delta \log P_{dio} + \kappa_a \Delta \log (w_o) \\
& + \kappa_s \Delta \log \left(1 + \frac{r_{sio} k_{sio}}{w_o l_{io}} \right) \\
& + \kappa_d \Delta \log \left(1 + \left(1 + \frac{r_{sio} k_{sio}}{w_o l_{io}} \right)^{-1} \times \frac{r_{dio} k_{dio}}{w_o l_{io}} \right) + X'_{io} \beta + \nu_{io}.
\end{aligned} \tag{18}$$

In this equation, the coefficients are defined as $\kappa_s = \frac{\sigma - \rho_s}{\rho_s - 1}$, $\kappa_d \equiv \frac{\sigma - \rho_d}{\rho_d - 1}$, and $\kappa_a \equiv -\sigma$. The term ν_{io} captures residual demand components, and X_{io} represents a vector of control variables.

Several endogeneity concerns arise for the estimation of this equation. First, our CEI measures, $\Delta \log P_{sio}$ and $\Delta \log P_{dio}$, can be correlated with the residual term ν_{io} if innovation activities respond endogenously to occupational demand shocks. For example, a surge in the demand for certain occupational services could incentivize firms to make more innovations in related capital goods. Second, since w_o is jointly determined with l_{io} , we need an exogenous shifter for labor supply. Lastly, capital-to-labor income ratios for task-similar and task-dissimilar capital can also be correlated with ν_{io} if demand shocks change capital intensity due to different elasticities of supply between labor and capital.

To address these concerns, we introduce five instrumental variables: two based on academic publications, two based on changes in import values, and one based on immigration shocks from Latin American countries. The publication and import instruments are constructed separately for task-similar and task-dissimilar capital. Our identification strategy rests on the exclusion restriction that initial exposures to these trends—academic publication growth, Latin American immigration, and capital imports—are predetermined and orthogonal to residual demand shocks at the occupation level, conditional on our set of controls. Appendix A.4 details the variation and relevance of the instruments.

We construct shift-share instruments for CEI that capture heterogeneous knowledge spillovers from academic publications to patents. For example, innovation in the com-

puter sector is based on knowledge produced in the field of electronic engineering. An increase in the number of papers in electronic engineering is positively correlated with innovation in the computer sector, but is unlikely to be correlated with demand shocks for IT workers. This IV approach is similar to that of [Aghion et al. \(2019\)](#) and [Berkes et al. \(2022\)](#), who use patent citations across regions and countries, respectively, to construct the share and use patent growth as a shifter. In contrast, we exploit citations from patents to academic publications to construct the share and use the growth rate of publications in European countries as the shifter to obtain more exogenous variation.

To measure the diffusion of knowledge from academic publications to patents, we use citation data from patents to academic publications as in [Marx and Fuegi \(2020\)](#); [Arora et al. \(2021\)](#).¹² We first construct the upstreamness of an academic field f to the patent class p by using citations made from 1970 to 1980.¹³ The upstreamness v_{pf} is calculated as below:

$$v_{pf} = \frac{C_{pf}}{\sum_f C_{pf}},$$

where C_{pf} is the number of citations from patent class p to academic field f .

We use this share to construct the exposure measure of the capital bundle k_{jio} to the academic field f . The upstream measure v_{pf} is multiplied by $s_{jiop}^{Pat.}$, the average share of patent class p in capital bundle k_{jio} , weighted by the expenditure-adjusted number of patents from 1970 to 1980.¹⁴

$$z_{jio}^{Pub.} = \sum_p s_{jiop}^{Pat.} \sum_f v_{pf} \Delta \log(\mathcal{P}_f), \quad (19)$$

where $\mathcal{P}_f$ is the number of publications. Papers affiliated only with European institutions are counted to avoid the potential bias from US patenting firms that also engage in academic projects, which could be correlated with residual demand shocks. The instrument

¹²A large number of citations from patents within a technology class to papers in a specific academic field suggests that the academic field serves as an upstream source of knowledge for that technology class. [Marx and Fuegi \(2020\)](#) show that 17.6% of USPTO patents cite at least one academic paper, with an average of two academic citations per patent.

¹³For the academic field, the Web of Science Field is used, encompassing 251 distinct fields. For patents, three-digit IPC patent classes, comprising 387 classes, are used.

¹⁴Appendix A.4.1 provides the formal definition of $s_{jiop}^{Pat.}$.

increases if the academic fields relevant to the capital bundle experience faster growth in publications.

The immigration instrument uses heterogeneous shares of immigrants from Latin American countries at the occupation and industry level interacted with the growth of Latin American immigration. Immigrants from Latin American countries likely possess comparative advantages that differ from those of US-born workers, influencing occupational labor supply in distinct ways. For each occupation, the heterogeneous exposure to immigration shocks is computed based on the share of Latin American workers in 1980. The Bartik immigration shock for occupation o is defined by the following equation.

$$z_{io}^{Latin} = s_o^{Latin} \Delta \log \left(\mathbf{L}^{Latin} - \mathbf{l}_o^{Latin} \right). \quad (20)$$

In the equation, s_o^{Latin} is the share of workers from Latin American countries in occupation o in 1980. $\mathbf{L}^{Latin}$ and $\mathbf{l}_o^{Latin}$ are the number of Latin American immigrants and the number of immigrant workers in occupation o . Workers in occupation o are subtracted from the calculation of the supply shock to rule out the effect of occupation-level shocks that led to more immigrants from Latin America.

Lastly, we use capital import data to construct a shifter for capital expenditure. Specifically, we calculate the log change in import value from 1980 to 2015 for each capital category n . Then, the import shock is defined as below:

$$z_{jio}^{Import} = \sum_{n \in \mathbb{N}_{jio}} s_{jion}^{Import} \Delta \log m_n, \quad (21)$$

where s_{jion}^{Import} represents the current-cost stock share of the capital good n for the capital bundle k_{jio} in 1980. m_n is the value of imports. We expect that an increase in the value of imports would be positively associated with an increase in capital stock from 1980 to 2015. For example, if a capital bundle relied more on sewing machines and its import increased significantly due to cheaper supplies from other countries, capital expenditure $r_{jio}k_{jio}$ would also increase if the substitution elasticity is greater than one. This global increase in commodity-level trade is plausibly exogenous to the demand shock at the

occupation level between 1980 and 2015.

Controls include the task-offshorability index at the occupation level developed in [Autor and Dorn \(2013\)](#), industry fixed effects, and the initial levels of the log wage bill in 1980. The offshorability index controls for the changes in demand for occupational services related to international trade. We also formulate a Bartik-style control for occupational services with industry-level wage growth interacted with industry composition at the occupation level. Lastly, 123 occupations that do not have any task-similar capital are excluded from the estimation.

We exploit the variation at the industry-by-occupation level for two reasons. First, industries are highly heterogeneous in terms of their capital composition. For example, the manufacturing industry works more with machinery than the service industry. Thus, our CEI measure of machine operators gives more weight to machine-related patents for the manufacturing industry than for the service industry. In [Appendix A.5](#), we decompose the variation of CEI measures and their instruments between occupations and industries. The results suggest that more than 20% of the variation in CEI-d results from industry-level variation within occupations. Second, the occupational composition varies significantly between industries. Thus, it is essential to control for heterogeneous trends across industries when evaluating the effect of CEI on labor demand.

4.2 Results

Table 3 presents the results of the first stage. All instrumental variables exhibit coefficients that align with the prior expectations. Specifically, immigration shocks are negatively correlated with wage changes, and publication shocks for each type of capital are positively correlated with the patent measure. Import shocks for each type of capital are positively associated with the corresponding capital income ratios. Each regression exhibits a high F-statistic, and the Cragg-Donald Wald F-statistic is 10.3, indicating that our instruments provide sufficient power for identification.

The estimates of structural parameters are summarized in Table 4.¹⁵ The elasticity

¹⁵Table A6 in [Appendix D.2](#) presents the results where tools are weighted by the inverse of their matching

Table 3: First-Stage Results

	Wage	Patent-s	Patent-d	Capital-s	Capital-d
Birth Place	-0.91 (0.01)	0.06 (0.02)	0.04 (0.03)	0.02 (0.00)	-0.00 (0.00)
Publication-s	-0.20 (0.06)	2.59 (0.09)	-1.09 (0.12)	0.04 (0.01)	0.02 (0.01)
Publication-d	-0.82 (0.06)	-0.01 (0.09)	1.83 (0.12)	-0.07 (0.01)	0.03 (0.01)
Import-s	0.17 (0.02)	0.18 (0.03)	-0.37 (0.04)	0.09 (0.00)	0.01 (0.00)
Import-d	0.11 (0.04)	0.22 (0.05)	0.23 (0.08)	-0.01 (0.01)	0.04 (0.01)
N	7,645	7,645	7,645	7,645	7,645
F-statistics	1117.67	172.22	85.98	194.42	76.24

Notes. This table presents the results of the first-stage regression in Equation (18). Standard errors are in parentheses. Immigration is the instrumental variable defined in Equation (20). Import-d and Import-s represent import shocks for task-dissimilar and task-similar capital, respectively, as characterized in Equation (21). Publication-d and Publication-s represent publication shocks for task-similar and task-dissimilar capital, respectively, as defined in Equation (19). Wage is $\Delta \log w_o$. Patent-s and Patent-d are $\Delta \log P_{sio}$, and $\Delta \log P_{dio}$, respectively. Capital-s is $\Delta \log \left(1 + \frac{r_{sio}k_{sio}}{w_o l_{io}} \right)$, and capital-d is $\Delta \log \left(1 + \left(1 + \frac{r_{sio}k_{sio}}{w_o l_{io}} \right)^{-1} \times \frac{r_{dio}k_{dio}}{w_o l_{io}} \right)$.

of substitution between task-dissimilar capital and labor is estimated at 1.2, whereas the elasticity between labor and task-similar capital is 1.6. Our results are comparable to the estimates in [Caunedo et al. \(2023\)](#), who assume a single elasticity of substitution between capital and labor for each occupation. Using time-series variation in birth rates and the supply of educated workers to construct the occupation-level supply shifter, they find that the elasticity ranges from 0.7 to 2.2. The substitution elasticity across occupational labor inputs, σ , is estimated to be close to that of the literature. In [Caunedo et al. \(2023\)](#), the value is calibrated at 1.3. This paper estimates the value of 1.6. This value is larger than the estimate in [Lee and Shin \(2017\)](#), 0.7, but smaller than the one in [Burstein et al. \(2019\)](#), 2.

Our marginally higher estimates are likely to result from dealing with a longer time horizon and cross-sectional variation for estimation. Our estimation deals with long-term

rates in Wikipedia for CEI estimation.

Table 4: Parameter Estimates

	ρ_s	ρ_d	σ	γ_p	γ_s	γ_d
Estimate	1.63	1.24	1.59	-0.72	-0.74	1.57
SE	(0.44)	(0.20)	(0.36)	(0.02)	(0.09)	(0.18)

Notes. This table shows the estimates and standard errors of the regression equations (17) and (18). ρ_s (ρ_d) is the elasticity of substitution between task-similar (task-dissimilar) capital and labor. σ is the elasticity of substitution between different occupational services. γ_p is the coefficient of CEI on user costs of capital. γ_s (γ_d) is the coefficient of CEI-s (-d) on occupational service demand shifter.

adjustments in the labor market over three decades and uses cross-sectional variation. We also construct labor and capital supply shifters with cross-sectional exposures to trends in immigration from Latin America and capital imports.

The estimate for σ is marginally smaller than ρ_s but larger than ρ_d . Due to the high standard error of ρ_s , it is hard to tell whether ρ_s is significantly greater than ρ_d or σ . However, ρ_d is less than σ at the significance level of 5%. As discussed in Section 2.3, these values imply that the scale effect of task-similar capital on boosting occupational labor demand is smaller than the substitution effect between labor and task-similar capital. As a result, a lower user cost of task-similar capital reduces the relative labor demand. In contrast, the scale effect of task-dissimilar capital is stronger than the substitution effect. If the occupation has a low intensity of task-similar capital, a lower user cost of task-dissimilar capital raises the relative labor demand.

Table 4 also presents the estimates for the coefficient of CEI measures on the user costs of capital and demand shifters for occupational labor. The estimate for γ_p is negative and large in magnitude.¹⁶ An increase in CEI by 1% reduces the user cost of capital by 0.72%. Along with the estimates of elasticities, this negative coefficient estimate implies that CEI-s leads to stronger substitution with capital than CEI-d within occupations. The estimate for γ_s is negative, and the estimate for γ_d is positive. Consequently, CEI-s decreases the demand for occupational labor, whereas CEI-d raises it, even after considering their effects on user costs. These effects are also quantitatively important. An increase in CEI-s

¹⁶Alternatively, we run the regression at the NIPA capital good and industry level. Then, the coefficient of CEI on the user costs of a capital good is estimated similarly at -0.85 with a standard deviation of 0.07.

by 1% reduces occupational service demand by 0.74%, and an increase in CEI-d by 1% increases occupational service demand by 1.57%.

When we estimate the occupational labor supply equation using CEI and import shocks as demand instruments, we estimate a labor supply elasticity of 1.5.¹⁷ By comparison, [Caunedo et al. \(2023\)](#) calibrate an elasticity of 0.3 at the annual frequency and using more aggregated occupational categories. Given that our analysis spans more than three decades and uses more detailed occupational codes, their estimate likely represents a lower bound.¹⁸ Accordingly, we use 1.5 as the baseline elasticity of occupational labor supply in the counterfactual exercises, while also reporting results with an elasticity of 0.3 in Section 5.1.

5 CEI, Elasticity of Substitution, and the Labor Share

We employ our framework to evaluate the extent to which CEI has driven the decline in the U.S. labor share since the late 1970s ([Elsby et al., 2013](#); [Karabarbounis and Neiman, 2014](#)). Because our occupation-level data do not include expenditures on structures and profits, we define the labor share specifically as labor expenditure relative to the sum of labor and capital expenditures. Using this definition, the labor share decreased from 85.5% to 82.8% between 1980 and 2015. In national accounts, the wage bill as a share of GDP decreased from 62.4% to 59.5% during the same period. If we assume the combined share of profit and structure expenditure to be constant 73% of GDP, 68% of the decrease in the aggregate labor share can be explained by the reduction of in the wage bill relative to equipment expenditure.

To analyze these shifts, we first express the aggregate labor share as the expenditure-

¹⁷Specifically, we estimate $\Delta \log L_o = \psi + \eta \Delta \log w_o + \epsilon_o$, where η is the labor supply elasticity across occupations. The standard error is 0.75.

¹⁸We have 293 occupations based on the Census Bureau's 1990 occupation codes. In contrast, [Caunedo et al. \(2023\)](#) use one-digit occupation codes.

weighted average of labor shares at the occupation-industry level.

$$\text{Labor Share} = \sum s_{io}^{Input} \text{labor share}_{io}. \quad (22)$$

In this equation, s_{io}^{Input} represents the share of occupation o and industry i in the total expenditure on production inputs.

This formulation allows us to decompose the change in the aggregate labor share into two primary components: a within-occupation margin, which captures labor share changes holding income shares constant, and a between-occupation margin, which reflects shifts in factor income shares holding internal labor shares constant. Our results indicate that the aggregate decline is almost entirely accounted for by a decline in within-occupation labor shares (2.8 percentage points), whereas the between-occupation margin contributed only a negligible increase (0.1 percentage points).

These declines in within-occupation labor shares are concentrated in occupations characterized by high labor costs relative to capital. As shown in Figure 2, occupations with higher relative labor costs in 1980 initially maintained lower labor shares in 1980 and subsequently experienced more substantial share declines through 2015.¹⁹

We show that CEI levels and their composition are systematically related to the relative costs of labor to capital in Figure 3. First, CEI-s is higher for occupations whose wages are disproportionately higher than the capital costs. Second, higher labor costs relative to capital imply lower CEI-d. For these occupations, the level of CEI-s is close to that of CEI-d. Most occupations have higher CEI-d than CEI-s because information equipment is likely task-dissimilar capital, and more patents were made on this capital in 1980–2015. In line with the finding in Section 4 that CEI-s is more substitutable with labor and reduces occupational labor demand, these patterns suggest that technical changes are directed toward disproportionately substituting capital for labor when wages are higher relative to user costs of capital. Our results also confirm that occupations are heterogeneously ex-

¹⁹The relative labor cost is $w_o / (r_{sio}(1 - \bar{w}_{dio}) + r_{dio}\bar{w}_{dio})$, where $\bar{w}_{dio} = k_{sio} / (k_{sio} + k_{dio})$. The user cost of capital is the rental price of the capital stock, equivalent to a million dollars in 1980 value; consequently, the relative labor cost measures the cost of hiring one worker relative to buying the capital stock worth a million dollars in 1980.

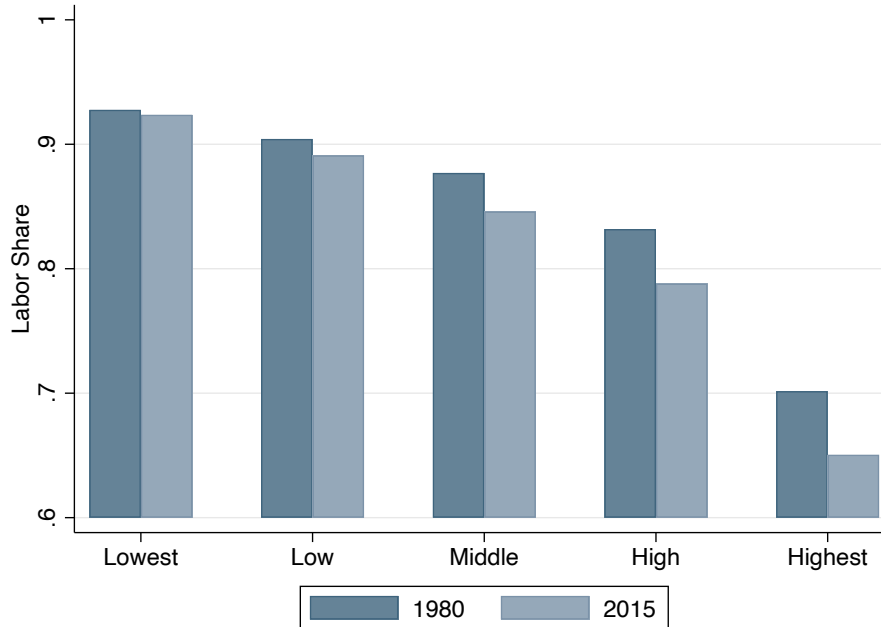


Figure 2: Relative Labor Cost and Labor Share in 1980 and 2015

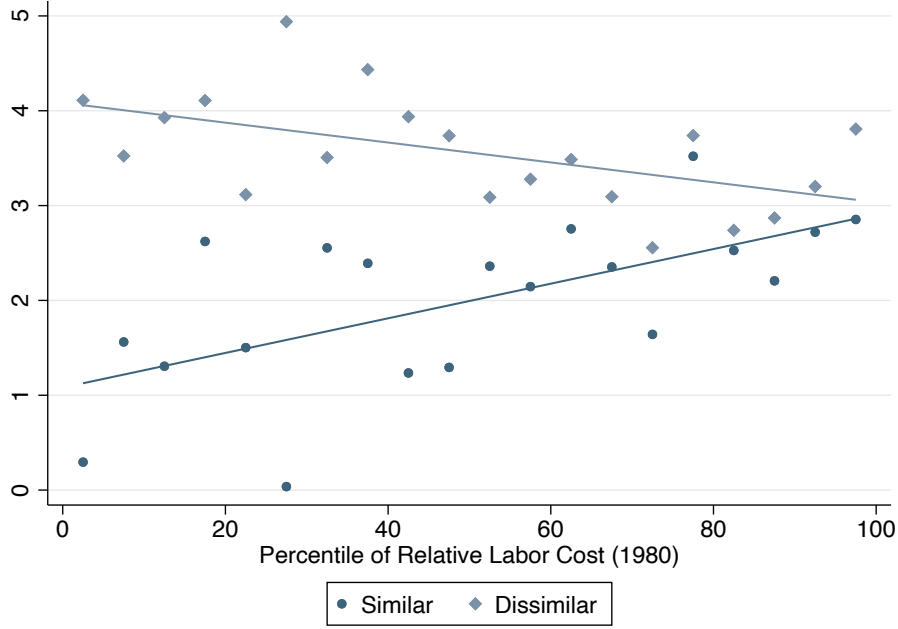
Notes: This graph shows the average within-occupation labor shares of equal-sized groups of occupations by their relative labor cost in 1980. Occupations are weighted by their factor income shares in 1980.

posed to innovations in magnitude and composition, consistent with the points made in [Autor et al. \(2003\)](#) and [Acemoglu and Restrepo \(2018a\)](#).

In our framework, CEI affects aggregate labor share in two ways. First, lower user costs of capital associated with CEI affect the aggregate labor share both within and between occupations. Because the elasticities of substitution of labor with both task-similar and task-dissimilar capital are greater than one, lower user costs decrease labor shares within occupations and industries. Furthermore, because the elasticity of substitution between occupational services also exceeds one and occupations vary in their capital intensities, heterogeneous changes in user costs shift their factor income shares. Given that occupations have different labor shares, these changes translate into changes in the aggregate labor share.

Second, changes in occupational labor demand directly associated with CEI also alter the aggregate labor share within and between occupations. Since CEI varies at the

Figure 3: CEI Measure across Relative Labor Cost



Notes. This figure presents the employment-weighted averages of CEI across three bins of occupations sorted by wage relative to user costs of capital in 1980. The CEI is defined in Equation (3).

occupation level, it directly reallocates labor demand and factor income shares between occupations. Because the supply elasticity is greater for capital than for labor, an increase in occupational labor demand raises wages relative to user costs and reduces within-occupation labor shares.

To quantify the effects of CEI on aggregate labor share, we conduct a counterfactual analysis by fixing CEI measures at their 1980 levels while holding all other 2015 demand and supply components constant.²⁰ We then describe the counterfactual and actual contributions of the changes in within-occupation labor shares and factor income shares to the aggregate labor share in Figure 4. As already pointed out, all five occupation groups experienced declines in labor share, and changes in factor income shares marginally con-

²⁰Appendix E.3 displays the effect of CEI on user costs of capital and how user costs alone can generate changes in the labor market and the aggregate labor share. Appendix E.4 presents the results when only computers are included in the CEI measure. In summary, the effects of computer-related patents on aggregate labor share are qualitatively consistent with the baseline but with a much smaller magnitude. User costs alone do not decrease but rather increase the aggregate labor share.

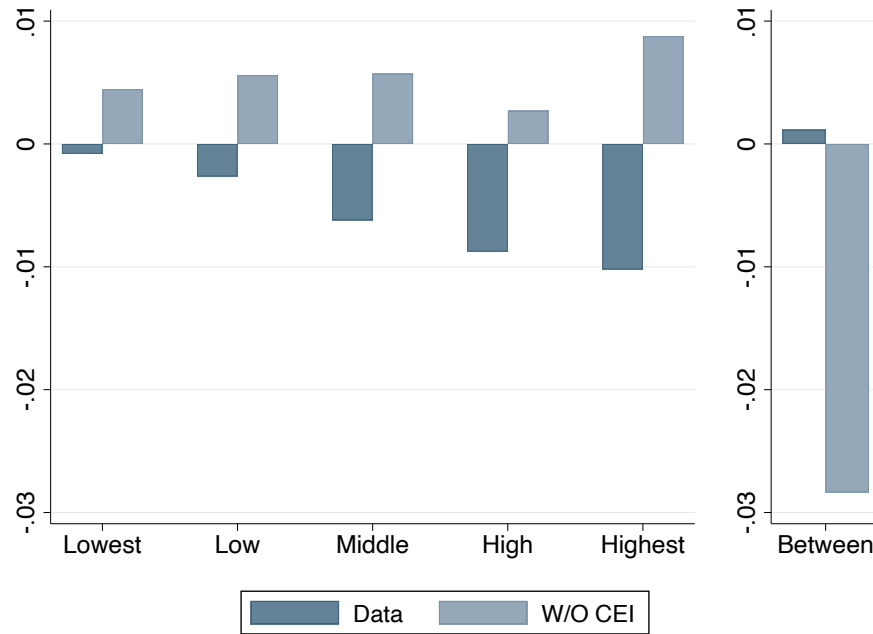


Figure 4: Role of CEI in Within-Occupation Labor Share Change

Notes: This graph decomposes the contribution of within-occupation changes in labor share. Each group corresponds to a quintile of wages divided by user costs of capital in 1980. Between-margin shows the contribution of factor share changes across occupations to the aggregate labor share. We set the elasticity of occupational labor supply at 1.5.

tributed to a smaller reduction in labor share. Without CEI, this pattern is completely reversed. Every occupation group would have experienced an increase in its internal labor share, and the between-occupation margin would have implied a reduction in the aggregate labor share. In summary, the model implies that the labor share would have barely changed without CEI, compared with the decrease of 2.7 percentage points in the data.

CEI exerts a disproportionately large impact on within-occupation labor shares for occupations characterized by high labor costs relative to capital. This pattern is primarily driven by the fact that higher relative labor costs are associated with significantly higher levels of CEI-s. Because task-similar capital possesses a higher elasticity of substitution with labor than task-dissimilar capital, the effect of CEI on internal labor shares is more pronounced when CEI lowers capital costs. In addition, since these high-labor-cost occu-



Figure 5: Similar Capital Share

Notes: This graph displays the binscatter plot between the share of task-similar capital and the relative labor cost in 1980. The dotted line shows the fitted values of the binscatter plot. The bars exhibit the share of occupations that have at least one task-similar capital good, where occupations are grouped by their relative labor cost in 1980. Occupations are weighted by their factor income shares.

pations exhibit a greater intensity of capital, a proportional change in user costs mechanically has a larger impact on labor shares.

At the same time, these occupations exhibit a higher intensity of task-similar capital, as illustrated in Figure 5. Occupations characterized by high relative labor costs are not only more likely to utilize task-similar capital but also to maintain a larger share of it within their capital bundles. This compositional shift increases the marginal impact of CEI-induced changes in user costs. As formally established in Equation (15), the occupation-level elasticity of substitution between capital and labor, $\hat{\rho}_{io}$, is an increasing function of the task-similar capital share. Figure 6 empirically confirms this relationship, showing that the substitution elasticity is significantly higher for occupations in the third and fifth quintiles compared to those in the first and second. Ultimately, the higher elasticity of substitution amplifies the impact of CEI, leading to a more pronounced contraction of within-occupation labor shares.

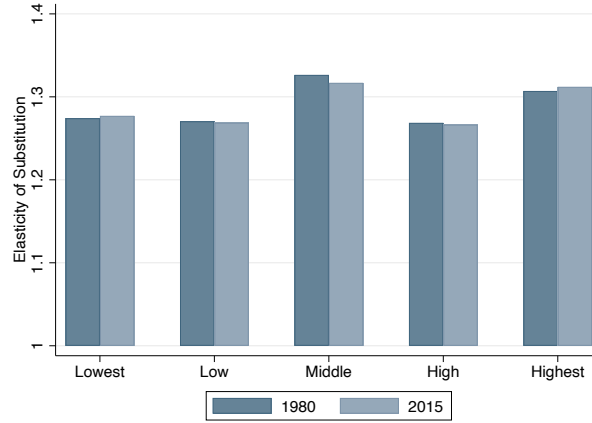


Figure 6: Elasticity of Substitution

Notes: This graph displays the occupation-level elasticity of substitution between labor and capital expressed in Equation (15). Each group corresponds to quintiles of wages divided by user costs of capital in 1980.

5.1 Discussion

Table 5 presents the counterfactual results across different values of labor supply elasticities and decomposes the aggregate changes into specific channels. Panel A demonstrates that our quantitative findings remain stable regardless of the specific value assigned to the elasticity of labor supply. This robustness stems from the fact that the estimated elasticity of substitution between occupations is near unity on the demand side. Consequently, factor income shares are relatively insensitive to labor supply shocks, as the price and quantity adjustments for labor effectively offset each other in terms of total expenditure shares. The degree to which CEI-driven reductions in capital user costs decrease within-occupation labor shares remains quantitatively invariant to the elasticity of labor supply.

The decomposition in Panel B reveals that both CEI-s and CEI-d put downward pressure on within-occupation labor shares.²¹ Notably, the effect of CEI-s is smaller because its magnitude is lower (see Figure 3), notwithstanding its significantly higher elasticity of substitution with labor. Conversely, on the between-occupation margin, CEI-d serves to

²¹Given that the same capital good can be task-similar for some occupations and task-dissimilar for others, this exercise aims to check whether capital contributed to the decline in aggregate labor share by executing the same functions as occupational labor or not.

mitigate the aggregate decline in the labor share; it is disproportionately concentrated in occupations with high initial labor shares (see Appendix Figure A8) and increases their factor income shares. Consequently, when accounting for the net effect across both margins, the total contribution of CEI-s to the decline in the labor share is larger than that of CEI-d.

Finally, Panel C distinguishes between the user cost and demand-shifter channels. The user cost channel serves as the primary driver of the aggregate labor share decline, operating through both the within- and between-occupation margins. Because the elasticities of substitution between labor and both task-similar and task-dissimilar capital exceed unity, a reduction in user costs fundamentally suppresses within-occupation labor shares. Furthermore, on the between-occupation margin, CEI increases the factor income shares of capital-intensive occupations, which typically possess lower baseline labor shares, thereby reducing the aggregate share. In contrast, the demand-shifter channel partially mitigates this decline, primarily through the between-occupation margin. Occupations with initially high labor shares tend to exhibit higher CEI-d and lower CEI-s, which increases their factor income shares. This effect is partially offset by the within-occupation margin, where higher labor demand reduces labor shares because the elasticity of labor supply is lower than that of capital supply.

6 Conclusion

This paper provides a new framework for quantifying the impact of CEI on the aggregate labor share by distinguishing between innovations that perform occupational tasks and those that complement them. We develop a novel methodology that matches more than 5.7 million patents to specific capital goods and occupational tasks using natural language processing. By classifying capital as task-similar or task-dissimilar based on its functional overlap with occupational tasks, we provide a systematic approach to characterizing the complex relationship between technological progress in capital and demand for human labor.

Table 5: Counterfactual Declines in Labor Share

	Within	Between	All
Actual Change	−2.79	0.11	−2.68
<u>Panel A. Varying Supply Elasticity</u>			
Elasticity=0.3	2.58	−2.45	0.14
Elasticity=1.5	2.48	−2.48	0.00
<u>Panel B. Similar vs. Dissimilar CEI</u>			
Similar	−0.79	−0.16	−0.95
Dissimilar	−0.24	−1.41	−1.66
<u>Panel C. Different Channels</u>			
Capital User Cost	2.28	1.23	3.51
Demand Shifter	−2.49	−4.79	−7.29

Notes. This table shows changes in the labor share between 1980 and 2015 in the actual data and counterfactual equilibria. The counterfactual equilibrium fixes the CEI measures at their levels in 1980. The column Within indicates changes in the labor share arising from changes in within-occupation labor shares, holding input income shares across occupations fixed, and the column Between shows changes in the labor share arising from changes in input income shares across occupations, holding within-occupation labor shares fixed. Panels B and C assume a labor supply elasticity of 1.5.

Our empirical results reveal a fundamental asymmetry in the way technology interacts with labor. We estimate that the elasticity of substitution between labor and task-similar capital is significantly higher than that between labor and task-dissimilar capital. These estimates imply that, while all forms of CEI reduce capital user costs, only those that mirror occupational tasks exert strong downward pressure on the aggregate labor share. We also find that innovation is empirically directed; CEI-s is disproportionately concentrated in occupations characterized by higher labor costs relative to capital, where the incentives for labor-displacing automation are greatest.

Our counterfactual analysis underscores the quantitative significance of these findings. We find that the observed decline in the U.S. labor share between 1980 and 2015 is almost entirely attributable to CEI. Innovation in task-similar capital serves as the primary driver of this result, owing to its high elasticity of substitution with labor and its concentration in occupations with initially low labor shares. Additionally, the downward pressure exerted by the user-cost channel on within-occupation labor shares is partially

moderated by the demand-shifter channel, which raises the factor income shares of occupations with high labor shares.

These findings suggest that the macroeconomic consequences of new technologies depend not only on the pace of innovation, but also on its direction. As recent AI and robotic technologies increasingly target functions previously considered non-substitutable, the distinction between task-similar and task-dissimilar capital will remain a critical lens for understanding the evolution of inequality and factor income shares. Our results call for future research that extends this framework to examine the implications of recent technological changes for ongoing macroeconomic trends.

References

- Acemoglu, Daron and Pascual Restrepo**, “Automation and new tasks: The implications of the task content of production for labor demand,” *Journal of Economic Perspectives*, 2018, 33 (2), 3–30.
- **and** —, “The Race Between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment,” *American Economic Review*, 2018, 108 (6), 1488–1542.
- **and** —, “Automation and new tasks: How technology displaces and reinstates labor,” *Journal of economic perspectives*, 2019, 33 (2), 3–30.
- **and** —, “Tasks, Automation, and the Rise in US Wage Inequality,” *Econometrica*, 2022, 90 (5), 1973–2016.
- Aghion, Philippe, Ufuk Akcigit, Antonin Bergeaud, Richard Blundell, and David Hémous**, “Innovation and Top Income Inequality,” *The Review of Economic Studies*, 2019, 86 (1), 1–45.
- Antras, Pol**, “Is the US aggregate production function Cobb-Douglas? New estimates of the elasticity of substitution,” 2004.
- Argente, David, Salomé Baslandze, Douglas Hanley, and Sara Moreira**, “Patents to Products: Product Innovation and Firm Dynamics,” *Working Paper*, 2023.
- Arora, Ashish, Sharon Belenzon, and Lia Sheer**, “Knowledge Spillovers and Corporate Investment in Scientific Research,” *American Economic Review*, 2021, 111 (3), 871–98.
- Autor, David, Caroline Chin, Anna Salomons, and Bryan Seegmiller**, “New Frontiers: The Origins and Content of New Work, 1940–2018,” *The Quarterly Journal of Economics*, 2024, p. qjae008.
- , **David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen**, “The fall of the labor share and the rise of superstar firms,” *The Quarterly journal of economics*, 2020, 135 (2), 645–709.

- Autor, David H and David Dorn**, “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market,” *American Economic Review*, 2013, 103 (5), 1553–1597.
- , **Frank Levy, and Richard J Murnane**, “The Skill Content of Recent Technological Change: An Empirical Exploration,” *The Quarterly Journal of Economics*, 2003, 118 (4), 1279–1333.
- , **Lawrence F Katz, and Melissa S Kearney**, “The Polarization of the US Labor Market,” *American Economic Review*, 2006, 96 (2), 189–194.
- Berkes, Enrico, Kristina Manyasheva, and Martí Mestieri**, “Global Innovation Spillovers and Productivity: Evidence from 100 Years of World Patent Data,” 2022.
- Berlingieri, G, D Lashkari, F Boeri, and J Vogel**, “Capital-Skill Complementarity in Firms and in the Aggregate Economy,” *Working Paper*, 2022.
- Bloom, Nicholas, Tarek Alexander Hassan, Aakash Kalyani, Josh Lerner, and Ahmed Tahoun**, “The Diffusion of Disruptive Technologies,” *NBER Working Paper*, 2021, (28999).
- Burstein, Ariel, Eduardo Morales, and Jonathan Vogel**, “Changes in Between-Group Inequality: Computers, Occupations, and International Trade,” *American Economic Journal: Macroeconomics*, 2019, 11 (2), 348–400.
- Bárany, Zsófia L and Christian Siegel**, “Job Polarization and Structural Change,” *American Economic Journal: Macroeconomics*, 2018, 10 (1), 57–89.
- Caunedo, Julieta, David Jaume, and Elisa Keller**, “Occupational Exposure to Capital-Embodied Technical Change,” *American Economic Review*, 2023, 113 (6), 1642–1685.
- Chirinko, Robert S and Debdulal Mallick**, “The substitution elasticity, factor shares, and the low-frequency panel model,” *American Economic Journal: Macroeconomics*, 2017, 9 (4), 225–253.
- Dierdorff, EC, DW Drewes, and JJ Norton**, “O*NET Tools and Technology: A Synopsis of Data Development Procedures,” 2006.

- Elsby, Michael WL, Bart Hobijn, and Ayşegül Şahin**, “The decline of the US labor share,” *Brookings papers on economic activity*, 2013, 2013 (2), 1–63.
- Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Annie Chen, Grace Cooper, Stephanie Richards, Megan Schouweiler, and Michael Westberry**, “IPUMS CPS: Version 11.0 [dataset],” 2023.
- Goldin, Claudia and Lawrence F Katz**, *The Race Between Education and Technology*, Harvard University Press, 2008.
- Goos, Maarten, Alan Manning, and Anna Salomons**, “Explaining Job Polarization: Routine-Biased Technological Change and Offshoring,” *American Economic Review*, 2014, 104 (8), 2509–26.
- and —, “Lousy and Lovely Jobs: The Rising Polarization of Work in Britain,” *The Review of Economics and Statistics*, 2007, 89 (1), 118–133.
- Griliches, Zvi**, “Capital-Skill Complementarity,” *The Review of Economics and Statistics*, 1969, pp. 465–468.
- Hémous, David, Morten Olsen, Carlo Zanella, and Antoine Dechezleprêtre**, “Induced Automation Innovation: Evidence from Firm-level Patent Data,” *Journal of Political Economy*, 2025, 133 (6).
- Hornstein, Andreas, Per Krusell, and Giovanni L Violante**, “The Effects of Technical Change on Labor Market Inequalities,” *Handbook of Economic Growth*, 2005, 1, 1275–1370.
- Hubmer, Joachim**, “The race between preferences and technology,” *Econometrica*, 2023, 91 (1), 227–261.
- Jorgenson, Dale W**, “Capital Theory and Investment Behavior,” *The American Economic Review*, 1963, 53 (2), 247–259.
- Karabarbounis, Loukas and Brent Neiman**, “The global decline of the labor share,” *The Quarterly journal of economics*, 2014, 129 (1), 61–103.

- Keller, Wolfgang and Håle Utar**, “International Trade and Job Polarization: Evidence at the Worker Level,” *Journal of International Economics*, 2023, 145, 103810.
- Kelly, Bryan, Dimitris Papanikolaou, Amit Seru, and Matt Taddy**, “Measuring Technological Innovation Over the Long Run,” *American Economic Review: Insights*, 2021, 3 (3), 303–20.
- Kogan, Leonid, Dimitris Papanikolaou, Lawrence Schmidt, and Bryan Seegmiller**, “Technology and Labor Displacement: Evidence from Linking Patents with Worker-Level Data,” *NBER Working Paper*, 2023, (w31846).
- Krusell, Per, Lee E Ohanian, José-Víctor Ríos-Rull, and Giovanni L Violante**, “Capital-Skill Complementarity and Inequality: A Macroeconomic Analysis,” *Econometrica*, 2000, 68 (5), 1029–1053.
- Lee, Sang Yoon Tim and Yongseok Shin**, “Horizontal and Vertical Polarization: Task-Specific Technological Change in a Multi-Sector Economy,” *NBER Working Paper*, 2017, (w23283).
- Mann, Katja and Lukas Püttmann**, “Benign Effects of Automation: New Evidence from Patent Texts,” *Review of Economics and Statistics*, 2023, 105 (3), 562–579.
- Marx, Matt and Aaron Fuegi**, “Reliance on Science: Worldwide Front-Page Patent Citations to Scientific Articles,” *Strategic Management Journal*, 2020, 41 (9), 1572–1594.
- Michaels, Guy, Ashwini Natraj, and John Van Reenen**, “Has ICT Polarized Skill Demand? Evidence from Eleven Countries over Twenty-Five Years,” *Review of Economics and Statistics*, 2014, 96 (1), 60–77.
- Oberfield, Ezra and Devesh Raval**, “Micro Data and Macro Technology,” *Econometrica*, 2021, 89 (2), 703–732.
- Ruggles, Steven, Sarah Flood, Matthew Sobek, Daniel Backman, Annie Chen, Grace Cooper, Stephanie Richards, Renae Rogers, and Megan Schouweiler**, “IPUMS USA: Version 14.0 [dataset],” 2023.

Sinha, Arnab, Zhihong Shen, Yang Song, Hao Ma, Darrin Eide, Bo-June Hsu, and Kuansan Wang, “An Overview of Microsoft Academic Service (MAS) and Applications,” in “Proceedings of the 24th International Conference on World Wide Web” 2015, pp. 243–246.

Webb, Michael, “The Impact of Artificial Intelligence on the Labor Market,” *Working Paper*, 2019.

Zhestkova, Yulia, “Technology Cluster Dynamics and Network Structure.” PhD dissertation, The University of Chicago 2021.

APPENDIX

(For Online Publication)

A Data Appendix

A.1 Data Construction

Our measure of CEI is constructed by integrating O*NET “Tools Used” data with patent records from the United States Patent and Trademark Office (USPTO). The O*NET data encompass 4,180 distinct capital goods utilized across 775 occupations, classified according to the 2010 Standard Occupational Classification (SOC) and the eight-digit United Nations Standard Products and Services Code (UNSPSC).

We utilize USPTO bulk files to identify 5.7 million utility patents with application years between 1970 and 2015, and the bulk files are downloaded from <https://patentsview.org>. To capture quality-improving innovations specifically, we exclude design and plant patents. Patent counts are attributed based on the application year rather than the grant year to more accurately reflect the timing of the underlying innovation. Functional descriptions for these capital goods are retrieved from Wikipedia articles, which serve as the textual basis for our similarity-matching algorithm.

Occupational employment and demographic data are sourced from the 1980 Decennial Census and the American Community Survey (ACS) for the 2015–2019 period. All employment figures are harmonized using the OCC1990 classification system provided by IPUMS. Our primary analytical sample is restricted to prime-age workers (25–54) to ensure labor market consistency across the sample period.

Occupational wages are derived from the Annual Social and Economic Supplement (CPS-ASEC). To isolate shifts in labor productivity from changes in worker composition, we calculate residualized wages using a Mincerian regression. Following [Berlingieri et al. \(2022\)](#), this model controls for age, education, race, and year fixed effects. The regression is estimated on a subsample of prime-age male workers, working 40+ weeks per year, to minimize potential selection bias. We further refine this sample by omitting observations with missing characteristics or nominal hourly wages below 50% of the federal minimum wage.

To align disparate data sources, we employ a series of systematic crosswalks. The 2010 SOC codes from O*NET are mapped to the OCC1990 system using correspondence tables from the 2012–2018 ACS. Similarly, we convert IND1990 industry variables into NAICS codes, which are subsequently aggregated to the 63 NIPA industries defined by the Bureau of Economic Analysis (BEA). This aggregation ensures that our occupation-level estimates of capital stocks and prices remain consistent with national accounting standards.

To construct our suite of instrumental variables, we integrate the following supplementary datasets. First, we use the Microsoft Academic Graph (MAG) from [Sinha et al. \(2015\)](#) to track publication volumes across 251 distinct scientific fields. To ensure the exogeneity of our knowledge-supply shifter, we count only those publications affiliated with European institutions. Second, following [Marx and Fuegi \(2020\)](#), we leverage patent citation data to quantify the knowledge flow from academic research to specific technological classes. Lastly, international trade data are sourced from UN Comtrade. We manually crosswalk SITC Rev. 2 (for 1980) and HS 1992 (for 2015) commodity codes to the corresponding NIPA capital categories to capture exogenous shifts in global capital supply.

A.2 Measures of Capital Stock and User Costs

This section details the methodology used to impute occupation-specific capital stocks and calculate their corresponding user costs, following the structural approach established by [Caunedo et al. \(2023\)](#).

Each occupation has a set of capital goods in the UNSPSC. These goods are converted to NIPA capital categories using the crosswalk in Table 1 of the Online Appendix for [Caunedo et al. \(2023\)](#). The 2012 fixed-price capital stock series is used to measure the quantity of capital. The capital intensity of an occupation o for the NIPA capital category n is first defined as the number of capital goods in the UNSPSC from “Tools Used” that are mapped into that category. Let $\#Capital_o^{s,n}$ ($\#Capital_o^{d,n}$) denote the number of task-similar (task-dissimilar) capital goods in the UNSPSC.

Using the fixed-cost capital stock (based on the fixed price in 2012 USD), K_{int} , we impute the occupation-industry capital stock as follows:

$$x_{siont} = \frac{l_{iot} \#Capital_o^{s,n}}{\sum_p l_{ipt} \#Capital_p^{s,n} + \sum_p l_{ipt} \#Capital_p^{d,n}} K_{int} \quad (A1)$$

$$x_{diont} = \frac{l_{iot} \#Capital_o^{d,n}}{\sum_p l_{ipt} \#Capital_p^{s,n} + \sum_p l_{ipt} \#Capital_p^{d,n}} K_{int}, \quad (A2)$$

where l_{iot} is the number of workers in industry i and occupation o at time t . Thus, capital stocks are prorated between occupations with an intensity-weighted number of workers. For occupations missing from the O*NET data, we assign the average tool intensity of the respective industry to adjust the capital stock, though these observations are excluded from the final regression analysis.

The price deflator, q_{int} , is calculated as the ratio between current-cost and fixed-cost capital stocks from the BEA and used as our capital price index. We derive physical depreciation rates (δ_{int}) by adjusting the BEA-reported depreciation (d_{int}) to account for

price changes in consumption (λ_t^c) and capital goods:

$$1 - \delta_{int} = (1 - d_{int}) \frac{q_{int}/\lambda_t^c}{q_{in,t-1}/\lambda_{t-1}^c} \quad (\text{A3})$$

Following [Jorgenson \(1963\)](#), the user cost for a single capital category n in industry i at year t is defined as:

$$\lambda_{int}^k = \frac{q_{int-1}}{\lambda_{t-1}^c} \left[R - (1 - \bar{\delta}_{int}) \frac{q_{int}/\lambda_t^c}{q_{int-1}/\lambda_{t-1}^c} \right]. \quad (\text{A4})$$

We set the gross return on a safe asset, R , to 1.03 and utilize the average physical depreciation rate ($\bar{\delta}_{int}$) for the corresponding decade. We use $\lambda_t^c = 1$.

We construct the aggregate capital bundle (k_{jiot}) for each occupation-industry pair using a chained index. The growth rate of the bundle, κ_{jiot}^k , is an expenditure-weighted average of the growth rates of individual NIPA categories (κ_{int}^k):

$$k_{jiot} = k_{jio1980} e^{\kappa_{jiot}^k}, \quad k_{jio1980} = \sum_n x_{jion1980} \quad (\text{A5})$$

$$\kappa_{jiot}^k = \sum_n \frac{\lambda_{int}^k x_{jiont}}{\sum_{n'} \lambda_{in't}^k x_{jion't}} \kappa_{int}^k. \quad (\text{A6})$$

In a departure from [Caunedo et al. \(2023\)](#)—who normalize user costs to unity across all occupation-industry pairs—we calibrate occupation-level capital stocks to their 1980 market values. This normalization implies that the price of capital bundles is set to one in the base year of 1980. We adopt this specification specifically to preserve the cross-sectional heterogeneity in capital stocks and user cost series at the occupation level.

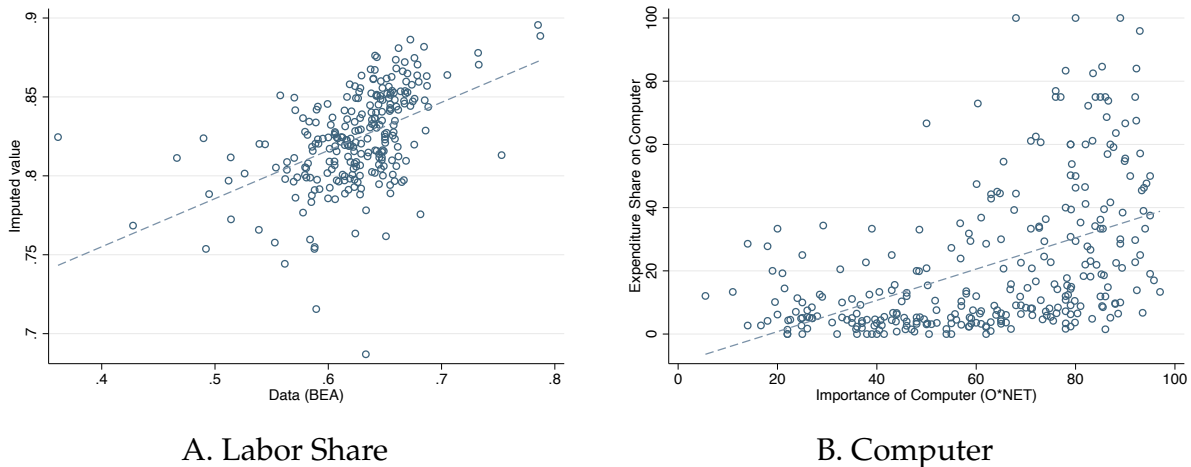
Finally, the user cost for the resulting capital bundle (r_{jiot}) is derived from the zero-profit condition.

$$r_{jiot} = \frac{\sum_n \lambda_{int}^k x_{jiont}}{k_{jiot}}. \quad (\text{A7})$$

A.3 Validation

To ensure the reliability of our occupation-level capital imputation, we perform two validation exercises that benchmark our results against external data sources and show the resulting scatter plots in Figure A1.

Figure A1: Validation Exercises



Notes. Panel A plots state-level labor shares, with the imputed values on the Y-axis and the corresponding BEA values on the X-axis, for both 1980 and 2015. Panel B plots the expenditure share of computers on the Y-axis against the O*NET measure of computer importance on the X-axis. The unit of observation is the occupation.

We first compute occupational labor shares, defined as labor expenditure relative to the total expenditure on labor and capital. These occupational shares are then aggregated to the state level using an expenditure-weighted average. We compare these imputed values with empirical state-level labor share data from the Bureau of Economic Analysis (BEA). Despite structural differences—specifically, that our occupational measure excludes corporate profits and expenditures on structures—the two series exhibit a robust correlation coefficient of 0.56 over the 1980–2015 period.

In the second exercise, we cross-validate our imputed computer expenditure shares against external metrics of technology use. Specifically, we correlate the O*NET “Importance of Computer” index with the expenditure shares imputed at the occupation level. This comparison yields a significant correlation of 0.48, further substantiating the accuracy of our imputation methodology.

A.4 Instrumental Variables

This appendix details the construction and identification logic of the five instrumental variables (IVs) used in our structural estimation.

A.4.1 Academic Publication Shock

We construct shift-share instruments that capture the diffusion of knowledge from academia to commercial patenting. The underlying logic is that scientific breakthroughs in an academic field (e.g., electronic engineering) accelerate innovation in downstream capital sectors (e.g., computer hardware), while being unlikely to respond endogenously to specific occupational demand shocks.

To quantify the “upstreamness” of an academic field f relative to a patent class p , we utilize citation data from patents to scientific articles. We calculate the citation share based on citations made between 1970 and 1980:

$$v_{pf} = \frac{C_{pf}}{\sum_f C_{pf}},$$

where C_{pf} is the number of citations from patent class p to academic field f . The left panel of Figure A2 plots v_{pf} , showing the variation in citation shares across patent classes. Engineering and chemistry are the fields that receive the most citations from patents.

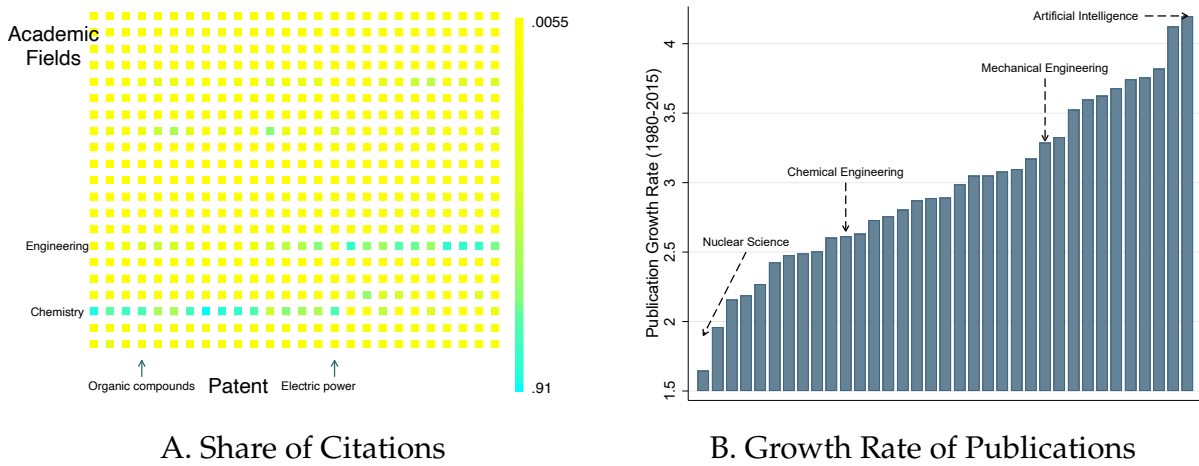
The exposure of a capital bundle k_{jio} to the academic field f is determined by $s_{jiop}^{Pat.}$, the share of patent class p in the capital bundle during the 1970–1980 period. Specifically, $s_{jiop}^{Pat.}$ is defined as follows:

$$s_{jiop}^{Pat.} = \sum_n \left(\frac{\lambda_{jion}^k x_{jion} \#Patent_{jon}}{\sum_{n'} \lambda_{jion'}^k x_{jion'} \#Patent_{jon'}} \sum_{u \in \mathbb{U}(j,o,n)} \frac{1}{\#Capital_o^{j,n}} \left(\frac{\widehat{\#Patent}_{up}}{\sum_{p'} \widehat{\#Patent}_{up'}} \right) \right), \quad (A8)$$

where $\frac{\lambda_{jion}^k x_{jion} \#Patent_{jon}}{\sum_{n'} \lambda_{jion'}^k x_{jion'} \#Patent_{jon'}}$ is the expenditure-adjusted patent share of the NIPA capital category n for the capital bundle k_{jio} . $\#Patent_{jon}$ is the number of patents matched with the capital good n for jo . $\mathbb{U}(j,o,n)$ is the set of UNSPSC u of type j used by occupation

o corresponding to n . $\#Capital_o^{j,n}$ is the number of capital goods in the UNSPSC that are classified as NIPA category n and is the cardinality of $\mathbb{U}(j, o, n)$. Lastly, $\widehat{\#Patent}_{up}$ is the number of patents in class p that are matched to the UNSPSC u . Thus, we give a higher weight to a patent class if more patents matched to the occupation belong to that patent class, the capital goods linked to the patent class are more representative in NIPA capital categories, and the NIPA category associated with the patent class accounts for a larger share of expenditures and the knowledge stock in the pre-period (1980).

Figure A2: Citation Share and Publication Growth Rate



Notes. Panel A plots the share of citations from patent technology classes (row) to academic fields (column) in 1970–1980. The graph only contains the IPC classes that have more than 50,000 citations to science in the entire period. When the color gets closer to blue, it has a higher citation share. Panel B displays the growth rates of publications between 1980 and 2015 in different academic fields. Publication data come from MAG and include publications associated only with European institutions.

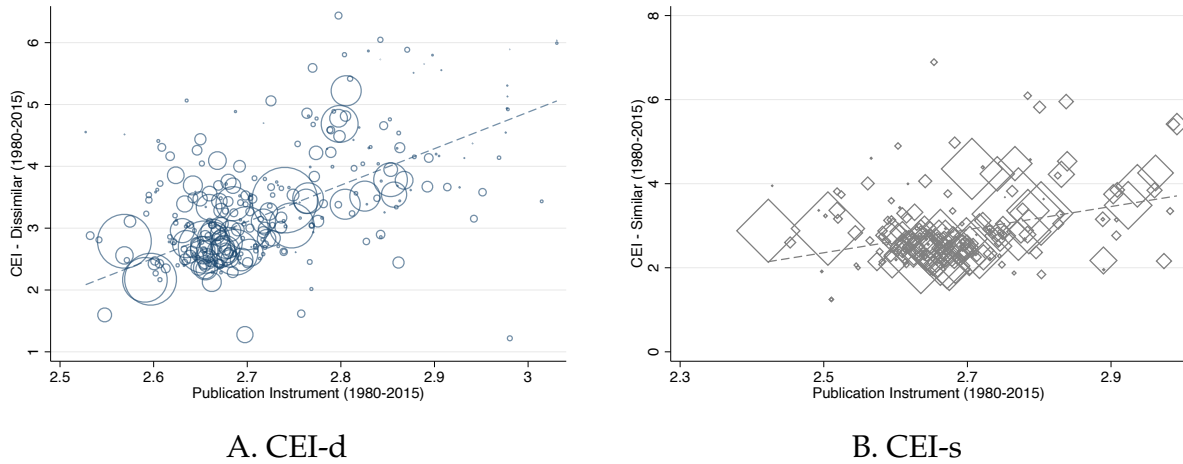
Finally, the publication instrument, $z_{jio}^{Pub.}$, is formally defined as:

$$z_{jio}^{Pub.} = \sum_p s_{jiop}^{Pat.} \sum_f v_{pf} \Delta \log(\mathcal{P}_f). \quad (A9)$$

To ensure exogeneity, $\Delta \log(\mathcal{P}_f)$ measures the growth rate of publications affiliated strictly with European institutions between 1980 and 2015. The right panel of Figure A2 displays the growth rates of publications in various academic fields. The fields with the highest growth rates include artificial intelligence, information systems, hardware, software engineering, and control systems.

Figure A3 displays the scatter plots between CEI measures and the resulting academic publication instruments, $z_{jio}^{Pub.}$, at the occupation level. The publication instruments are positively associated with the CEI measures, both for task-similar and task-dissimilar capital.

Figure A3: CEI and Publication Instrument



Notes. Panel A plots CEI-d, and Panel B plots CEI-s over the publication instrument. Each circle corresponds to an occupation code in OCC1990, and the size of a circle corresponds to the stock value of capital in 1980. CEI is residualized with industry fixed effects.

A.4.2 Immigration Shock

The immigration instrument identifies exogenous shifts in occupational labor supply by exploiting historical patterns of Latin American immigration.²²

Following a Bartik-style structure, the instrument interacts the 1980 share of Latin American workers in occupation o (s_o^{Latin}) with the aggregate growth of Latin American immigration. To eliminate the risk of occupation-specific shocks driving the aggregate flow, we apply a "leave-one-out" correction

$$z_{io}^{Latin} = s_o^{Latin} \Delta \log \left(\mathbf{L}^{Latin} - \mathbf{l}_o^{Latin} \right),$$

where $\mathbf{L}^{Latin}$ is the total number of Latin American immigrants and $\mathbf{l}_o^{Latin}$ is the number of immigrant workers in occupation o . This correction ensures that the supply shifter is not contaminated by shocks to the specific occupation being studied.

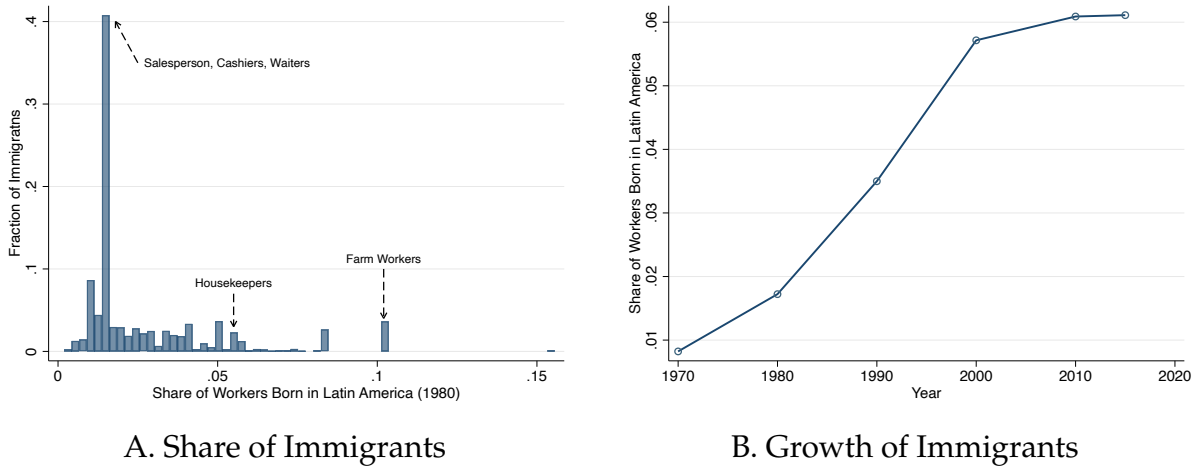
From 1980 to 2015, the population of Latin American workers in the US surged almost tenfold, while the number of US-born workers doubled. As a result, the share of workers born in Latin America in the total employment of the United States increased from 1.8% in 1980 to 6% in 2015, as shown in Panel B of Figure A4.

Immigrants from Latin America are likely to have comparative advantages different from those of US-born workers, influencing their choice of occupation differently. Panel A of Figure A4 shows the histogram of the share of workers from Latin America in 1980 for different occupations, with the weight of each occupation based on its employment numbers that year. The proportion of Latin American workers differs significantly between occupations. For example, in 1980, 10.2% of the farm workers were from Latin America, while 0.2% of therapists were born in the region. Consequently, a surge in Latin American immigration would disproportionately affect the labor supply in certain occupations, such as farm workers.

Figure A5 plots the immigration instrument with the wage and employment growth

²²We exclude workers from South America because they account for only 0.3% and 1% of workers in 1980 and 2015, respectively.

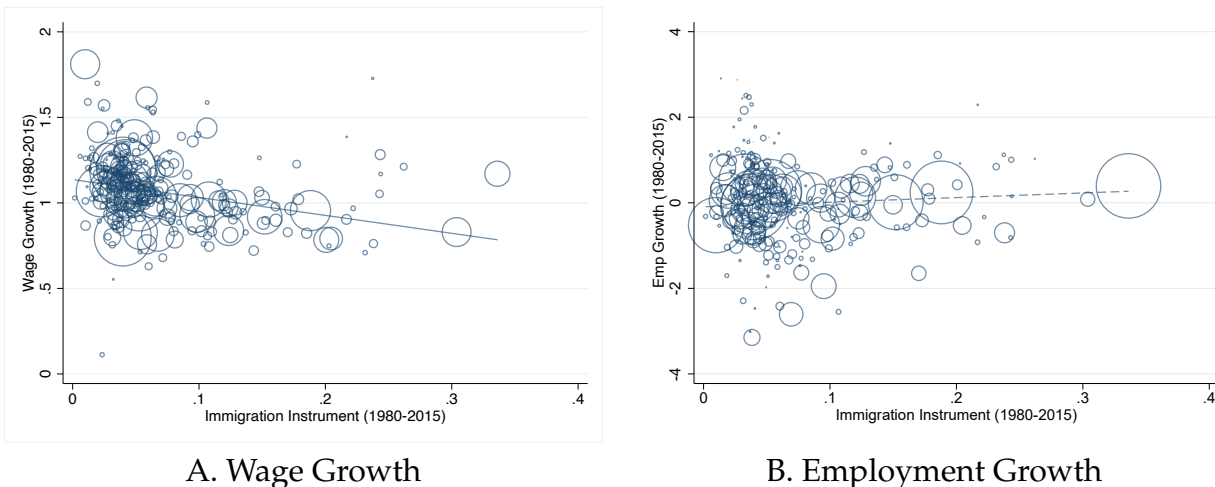
Figure A4: Share of Workers Born in Latin America



Notes. Panel A plots the share of workers born in Latin America in 1980 at the occupation level and draws the histogram of the observations. Each occupation is weighted by the number of immigrant workers in 1980. Panel B plots the share of workers in the US who were born in Latin America over time.

rates at the occupation level. The immigration instrument is negatively correlated with wage growth, while it is positively associated with employment growth. This suggests that our instrument serves as a shifter for labor supply.

Figure A5: Wage Change and Immigration Instrument



Notes. This figure plots the wage and employment growth rates of occupations between 1970 and 2015 with an immigration instrument. Each circle corresponds to an occupation code in OCC1990, and the size of a circle corresponds to the capital stock in 1980. Wage and employment growth are residualized with industry fixed effects.

A.4.3 Import Shock

We utilize capital import data to capture exogenous shifts in the supply and user cost of capital. An increase in global imports of a specific capital good (e.g., industrial sewing machines) likely indicates a fall in production costs or an expansion in global supply.

The import shock is defined as the expenditure-weighted average growth of import values across NIPA capital categories:

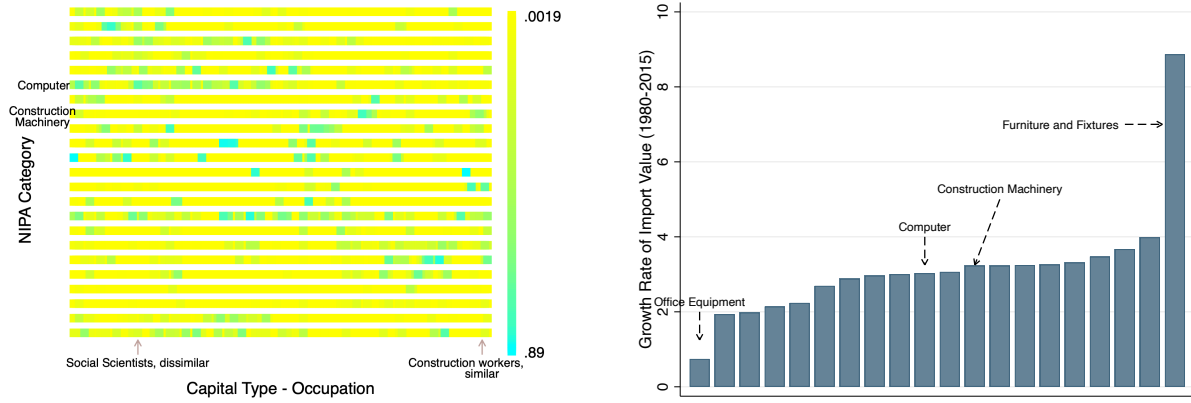
$$z_{jio}^{Import} = \sum_{n \in N_{jio}} s_{jion}^{Import} \Delta \log m_n ,$$

In this specification, s_{jion}^{Import} is the 1980 current-cost stock share of capital good n in the occupation-specific bundle k_{jio} , and m_n is the total value of U.S. imports for that category.

The capital expenditure share in 1980 serves as an exposure measure, while the growth rate of import value is used as a shifter. The growth rate of import value is sourced from Comtrade data from 1980 to 2015. We use data on imports from all countries to the U.S. at the four-digit SITC Rev. 2 level. The SITC codes are manually matched to the NIPA capital categories. Table A1 displays the mapping.

The left panel of Figure A6 displays the variation in the capital expenditure shares, and the right panel shows the distribution of growth rates of imported capital goods at the SITC Rev. 2 level. These figures suggest that the import instruments have sufficient variation both in the share and the shifter. Figure A7 plots the relationship between the import instruments and capital expenditure growth rates at the occupation level. Capital expenditure is calculated as the product of the stock and the user cost of capital goods. These results suggest that the capital import instrument is positively associated with capital expenditure, satisfying the relevance condition for the instrument.

Figure A6: Import Instrument and Capital Expenditure

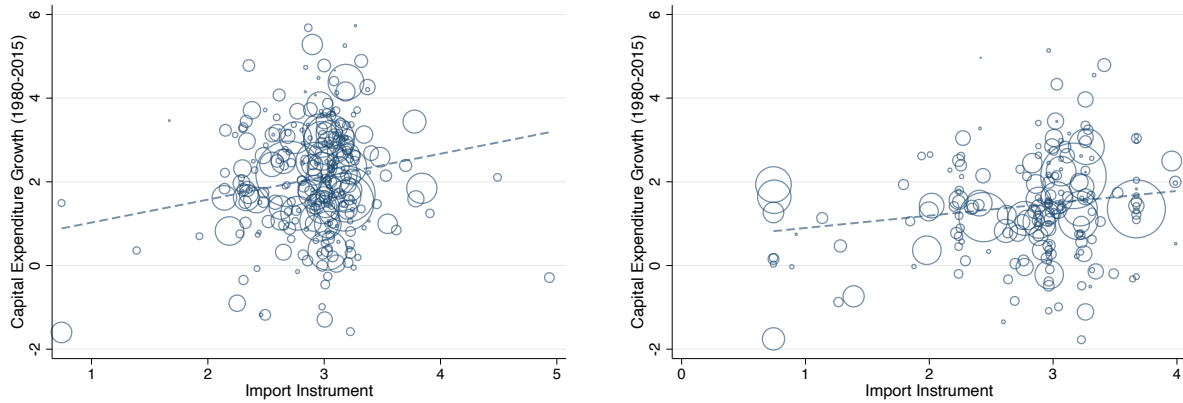


A. Capital expenditure share in 1980

B. Growth rate of import value

Notes. Panel A shows the capital expenditure share in 1980 across two-digit occupation codes by capital type on the X-axis and NIPA capital category on the Y-axis. Capital expenditure is calculated as the stock value multiplied by the user cost of the capital. Panel B displays the distribution of growth rates for the import value of capital goods from 1980 to 2015. The data are sourced from Comtrade, where we aggregate four-digit SITC product codes into NIPA categories.

Figure A7: Import Instrument and Capital Expenditure



A. Task-dissimilar capital

B. Task-similar capital

Notes. Panel A plots import instruments for task-dissimilar capital, and Panel B for task-similar capital over capital expenditure. Capital expenditure is stock values multiplied by user costs of capital. Each circle corresponds to an occupation code in OCC1990, and the size of a circle corresponds to the stock value of capital in 1980. The capital expenditure growth is residualized with industry fixed effects.

Table A1: Mapping Between NIPA Categories and HS Codes

NIPA code	Description of NIPA code	HS code	Description of HS code
4	Computers and peripheral equipment	8471	Automatic data processing machines and units thereof
4	Computers and peripheral equipment	8473	Parts for use with data processing machines
5	Communication equipment	8517	Line telephony and telegraphy apparatus, including telephones
5	Communication equipment	8525	Transmission apparatus for radio-telephony or broadcasting
5	Communication equipment	8526	Radar, radio navigational aid, and radio remote control apparatus
5	Communication equipment	8529	Parts for transmission and reception apparatus
6	Medical equipment and instruments	9018	Instruments for medical, surgical, dental, or veterinary use
6	Medical equipment and instruments	9019	Mechano-therapy, massage, and psychological aptitude-testing apparatus
6	Medical equipment and instruments	9021	Orthopedic appliances, including crutches, splints, and braces
6	Medical equipment and instruments	9022	X-ray apparatus and other medical diagnostic imaging apparatus
9	Nonmedical instruments	9031	Instruments for measuring or checking geometric quantities
9	Nonmedical instruments	9026	Instruments for measuring or checking liquid or gas flow, level, or pressure
9	Nonmedical instruments	9027	Instruments for physical or chemical analysis
9	Nonmedical instruments	9032	Automatic regulating or controlling instruments
10	Photocopy and related equipment	8443	Printing machinery and machines for uses ancillary to printing
10	Photocopy and related equipment	8442	Machinery and apparatus for printing plate preparation
11	Office and accounting equipment	8470	Calculating machines; accounting and ticketing machines
11	Office and accounting equipment	8472	Other office machines
13	Fabricated metal products	7308	Structures and parts of structures of iron or steel
13	Fabricated metal products	7326	Other articles of iron or steel
13	Fabricated metal products	7616	Other articles of aluminum
13	Fabricated metal products	7419	Other articles of copper
14	Engines and turbines	8411	Turbojets, turbopropellers, and other gas turbines
14	Engines and turbines	8406	Steam turbines and other vapor turbines
14	Engines and turbines	8407	Spark-ignition reciprocating or rotary internal combustion engines
14	Engines and turbines	8408	Compression-ignition internal combustion piston engines
17	Metalworking machinery	8456	Machine tools for working any material by removal of material
17	Metalworking machinery	8461	Machine tools for planing, shaping, slotting, and broaching
17	Metalworking machinery	8462	Machine tools for working metal by forging, hammering, or die-stamping
17	Metalworking machinery	8463	Other machine tools for working metals or carbides
18	General industrial equipment	8479	Machines and mechanical appliances with individual functions
18	General industrial equipment	8428	Lifting, handling, loading, or unloading machinery
18	General industrial equipment	8431	Parts suitable for use with lifting or moving machinery
20	Electrical equipment	8535	Electrical apparatus for switching electrical circuits
20	Electrical equipment	8536	Electrical apparatus for switching or protecting electrical circuits
20	Electrical equipment	8544	Insulated wire, cable, and other electric conductors
20	Electrical equipment	8504	Electrical transformers, static converters, and inductors
22	Trucks, buses, and truck trailers / Autos	8703	Motor cars and other motor vehicles for transporting people
22	Trucks, buses, and truck trailers / Autos	8704	Motor vehicles for the transport of goods
22	Trucks, buses, and truck trailers / Autos	8716	Trailers and semi-trailers; other vehicles
22	Trucks, buses, and truck trailers / Autos	8706	Chassis fitted with engines for motor vehicles
26	Aircraft	8802	Aircraft, including helicopters and airplanes
26	Aircraft	8803	Parts of aircraft or spacecraft
27	Ships and boats	8901	Vessels for transport of persons or goods
27	Ships and boats	8903	Yachts and other vessels for pleasure or sports
27	Ships and boats	8904	Tugs and pusher craft
28	Railroad equipment	8601	Rail locomotives powered by external sources
28	Railroad equipment	8602	Rail locomotives powered by an internal combustion engine
28	Railroad equipment	8607	Parts for railway or tramway locomotives
30	Furniture and fixtures	9401	Seats (except barber, dental, or similar chairs)
30	Furniture and fixtures	9403	Other furniture and parts thereof
33	Agricultural machinery	8432	Agricultural, horticultural, or forestry machinery
33	Agricultural machinery	8433	Harvesting, threshing, and other agricultural machines
33	Agricultural machinery	8436	Other agricultural, horticultural, forestry machinery
36	Construction machinery	8429	Self-propelled bulldozers, excavators, and road rollers
36	Construction machinery	8430	Other moving, grading, leveling, scraping, and boring machinery
39	Mining and oilfield machinery	8430	Other moving, grading, leveling, scraping, and boring machinery
40	Service industry machinery	8476	Automatic goods vending machines
40	Service industry machinery	8451	Machinery for laundering, drying, cleaning, or ironing textiles
88	Software	8523	Discs, tapes, and other recorded media
88	Software	8524	Recorded media for sound or video reproduction

Notes. This table displays the manual mapping between NIPA capital categories and HS codes (1992).

A.5 Industry Variation in CEI

We report the results of the variance decomposition of CEI measures between occupations and industries in Table A2. The variation in CEI-s predominantly comes from occupation-level heterogeneity in the lists of capital goods. However, a non-negligible fraction of variation comes from industry-level heterogeneity in capital composition within each occupation for CEI-d. Due to industry-level heterogeneity in occupational composition, between-industry variation in CEI measures are also sizeable in the second panel.

Table A2: Variance Decomposition

	Similar		Dissimilar	
	CEI	IV	CEI	IV
Between Occupation	0.906	0.967	0.803	0.873
Within Occupation	0.094	0.033	0.197	0.127
#Occupations	211	-	291	-
Between Industry	0.332	0.566	0.381	0.477
Within Industry	0.668	0.434	0.619	0.523
#Industries	59	-	59	-

Notes. This table shows the variation in the CEI measures and their instruments between occupations and industries.

A.6 CEI and Labor Share in 1980

Figure A8 illustrates the systematic relationship between the initial labor share in 1980 and the magnitude and composition of CEI between 1980 and 2015. Specifically, CEI-s is negatively correlated with the initial labor share, while CEI-d exhibits a positive correlation with the labor share in 1980.

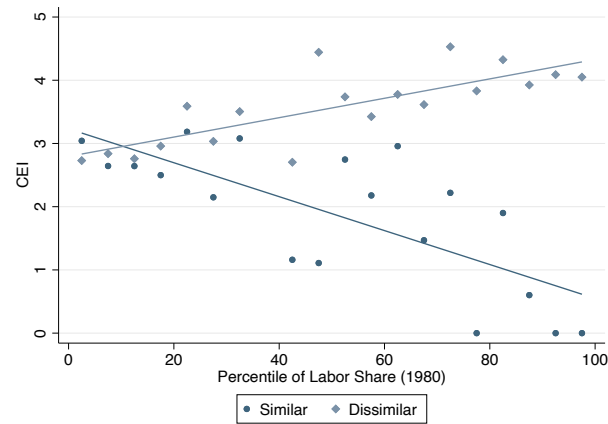


Figure A8: CEI and Labor Share in 1980

B Details in Text Matching

B.1 Share of Matched Tools and Patents

Table A3 summarizes the share of these tools successfully matched to functional descriptions on Wikipedia. We find significant variation in coverage across NIPA categories. Electronics, furniture, and machinery are more likely to be found in Wikipedia, while mining, medical equipment, and aircraft are less likely to be found.

Table A3: Share of Tools Found in Wikipedia

NIPA	Description	Found in Wikipedia (%)
20	Electrical transmission, distribution, and industrial apparatus	73.08%
4	Computers and peripheral equipment	69.64%
30	Furniture and fixtures	63.16%
27	Ships and boats	62.86%
40	Service industry machinery	60.00%
11	Office and accounting equipment	54.22%
29	Other equipment	53.13%
41	Electrical equipment, n.e.c.	53.10%
19	General industrial including materials handling equipment	52.03%
13	Fabricated metal products	49.62%
5	Communication equipment	48.59%
22,25	Trucks, buses, and truck trailers + autos	48.57%
14	Engines and turbines	46.81%
36	Construction machinery	44.44%
33	Agricultural machinery	42.86%
9	Nonmedical instruments	40.19%
10	Photocopy and related equipment	37.66%
18	Special industry machinery, n.e.c.	35.29%
39	Mining and oilfield machinery	31.13%
17	Metalworking machinery	30.61%
28	Railroad equipment	30.00%
6	Medical equipment and instruments	26.07%
26	Aircrafts	14.29%

Table A4 shows the share of patents matched to at least one tool, broken down by IPC patent class and time period. Our results indicate that the matching performance is highly dependent on the nature of the technology. Chemistry patents are rarely matched to tools, whereas engineering patents have higher matching rates than others.

Table A4: Patent-Tool Matching Rates across Patent Class and Period

Patent Class	Matching Rate (%)			
	1970–1980	1980–1990	1990–2000	2000–2015
Human necessities	22.50	22.07	22.69	18.38
Transportation	33.19	32.75	33.48	26.80
Chemistry	6.74	7.31	8.18	9.06
Textile	28.53	30.61	31.36	26.19
Construction	32.14	31.30	31.87	24.29
Engineering	43.23	42.47	42.99	34.49
Physics	28.06	27.60	25.50	20.86
Electricity	27.48	27.74	25.87	21.87

Notes. This table presents the share of patents matched with at least one tool by period and patent class (IPC one-digit level).

B.2 Cleaning text data

We apply techniques to ensure the integrity of the matching algorithm. First, we remove stopwords—common words such as “is,” “where,” and “have”—which are considered insignificant for delivering core technical content. Eliminating these terms prevents the algorithm from generating erroneous matches based on shared functional words rather than substantive descriptions. Following this step, the remaining words are lemmatized to standardize various grammatical forms into their base roots; for instance, terms like “generating” or “generated” are converted to “generate”. This standardization is essential to ensure that words with analogous meanings align appropriately across different documents, regardless of their original tense or plurality.

B.3 Examples

This section provides specific illustrations of how the text-matching algorithm identifies functional overlaps between patent documentation and capital goods descriptions. Example 1 demonstrates the direct semantic alignment between a patent abstract and a Wikipedia article, specifically for a protein sequencer. The presence of common technical terms—such as “amino acid,” “peptide,” and “protein identification”—allows the algorithm to generate a high similarity score by transforming these shared substantive words into weighted vectors.

Example 1: Example of Text Matching between Patent and Wikipedia

Patent: Systems, apparatuses and methods for reading an amino acid sequence (10139417)	Wikipedia: Protein sequencer
system apparatus method reading amino acid sequence embodiment present disclosure relate amino acid modified amino acid peptide protein identification sequencing mean example electronic detection individual amino acid small peptide	protein sequencing practical process determining amino acid sequence part protein peptide may serve identify protein characterize post translational modification typically partial sequencing protein provides sufficient information one sequence tag identify reference database protein sequence derived conceptual translation gene

Notes. This figure shows an example of an abstract of the matched patent and Wikipedia article of the capital good. The blue text is the common words between two texts.

Furthermore, Example 2 offers a broader selection of successfully matched capital goods and their corresponding patent titles. The description of the capital goods aligns with the titles of the matched patents.

Example 2: Example of Matched Capital Goods and Title of Patents

Capital Goods	Title of Patent
Battery chargers	Power tool, battery, charger and method of operating the same
Belt conveyors	Conveyor belt assembly
Cash registers	Theft proof cash drawer assembly
Desktop computers	Method and system for managing windows desktops in a heterogeneous server environment
Glass cutters	Discrete glass sheet cutting
Satellite phone	Communication system with direct link to satellite
Sewing machine needles	Multiple-needle sewing machine
Smoke detectors	Smoke detector system for a house
Tire pressure gauge	Tire pressure control system, tire pressure control device and tire pressure control method
Touch screen monitors	Technologies for interacting with computing devices using haptic manipulation

Notes. This table shows examples of capital goods and the titles of patents matched to them.

C Model Details

C.1 Micro-foundation of the Nested CES Production

We provide a micro-foundation for the nested CES production function structure described in Equations (4)–(6) by adapting the task-based framework established in [Acemoglu and Restrepo \(2018a\)](#).

Industrial output Y_i is a CES composite of occupational services $\mathbf{y}_{io}$, which are themselves produced by aggregating a continuum of occupational tasks $\mathbf{y}_{iov}$ in $v \in [0, V]$. Tasks within the range

$$[0, I]$$

are automated and performed exclusively by task-similar capital, while tasks in the range $(I, V]$ are implemented through a combination of occupational labor and task-dissimilar capital. The elasticity of substitution between occupational tasks is ρ_s . For each task, the elasticity of substitution between labor and task-dissimilar capital is given by ρ_d .

$$Y_i = \left(\sum_o \mathbf{y}_{io}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (\text{A10})$$

$$\mathbf{y}_{io} = \left(\int_0^V \mathbf{y}_{iov}^{\frac{\rho_s-1}{\rho_s}} dv \right)^{\frac{\rho_s}{\rho_s-1}}, \quad (\text{A11})$$

$$\mathbf{y}_{iov} = \begin{cases} \gamma_{iov}^s \tilde{z}_{sio}^{\frac{1}{\rho_s}} k_{siov} & \text{if } v \in [0, I], \\ \gamma_{iov}^l \left(l_{iov}^{\frac{\rho_d-1}{\rho_d}} + z_{dio}^{\frac{\rho_d-1}{\rho_d}} k_{dio}^{\frac{\rho_d-1}{\rho_d}} \right)^{\frac{\rho_d}{\rho_d-1}} & \text{if } v \in (I, V]. \end{cases} \quad (\text{A12})$$

For occupational labor input l_{io} and task-dissimilar capital k_{dio} , optimal allocation be-

tween tasks in $(I, V]$ implies

$$\begin{aligned} l_{iov} &\propto (\gamma_{iov}^l)^{\rho_s-1}, \\ l_{iov} &= l_{io} \times \frac{(\gamma_{iov}^l)^{\rho_s-1}}{\int_I^V (\gamma_{iov'}^l)^{\rho_s-1} dv'}, \\ k_{dio} &= k_{dio} \times \frac{(\gamma_{iov}^l)^{\rho_s-1}}{\int_I^V (\gamma_{iov'}^l)^{\rho_s-1} dv'}. \end{aligned} \quad (13)$$

$$\mathbf{y}_{io} = (\Gamma_{io}^l)^{\frac{\rho_s}{\rho_s-1}} \left(\left(l_{io}^{\frac{\rho_d-1}{\rho_d}} + z_{dio}^{\frac{1}{\rho_d}} k_{dio}^{\frac{\rho_d-1}{\rho_d}} \right)^{\frac{\rho_d}{\rho_d-1} \frac{\rho_s-1}{\rho_s}} + \frac{\Gamma_{io}^s}{\Gamma_{io}^l} \tilde{z}_{sio}^{\frac{1}{\rho_s}} (k_{sio})^{\frac{\rho_s-1}{\rho_s}} \right)^{\frac{\rho_s}{\rho_s-1}}, \quad (A14)$$

$$\Gamma_{io}^l = \left(\int_I^V (\gamma_{iov}^l)^{\rho_s-1} dv \right)^{\frac{1}{\rho_s}}, \quad (A15)$$

$$\Gamma_{io}^s = \left(\int_0^V (\gamma_{iov}^k)^{\rho_s-1} dv \right)^{\frac{1}{\rho_s}}. \quad (A16)$$

We can establish the following equivalence results: $y_{io} = \mathbf{y}_{io} / (\Gamma_{io}^l)^{\frac{\rho_s}{\rho_s-1}}$. $\mu_{io} = (\Gamma_{io}^l)^{\frac{\rho_s}{\rho_s-1}}$, and $z_{sio} = (\tilde{z}_{sio} \Gamma_{io}^s / \Gamma_{io}^l)^{\frac{1}{\rho_s}}$. When CEI-s expands the range of automated tasks I , it reduces μ_{io} . On the other hand, if CEI-d expands V through the creation of new and complex tasks, it increases μ_{io} .

This interpretation implies that z_{sio} increases in CEI-s but decreases in CEI-d. We run a regression of $\Delta \log z_{sio}$ on CEI-s and CEI-d instrumented with the publication and import instruments, along with industry fixed effects. The regression yields the following results.

$$\Delta \log \hat{z}_{sio} = \underset{(0.21)}{-0.35} \Delta \log P_{dio} + \underset{(0.13)}{3.26} \Delta \log P_{sio} + d_i. \quad (17)$$

The numbers in parentheses are the standard errors. These numbers confirm our predictions. Even when CEI-d and CEI-s are additionally allowed to affect $\Delta \log z_{sio}$, the estimation results are the same because Equation (18) remains unchanged. However, the

counterfactual change in labor share is affected by CEI altering demand for capital for fixed user costs. In this case, the aggregate labor share now increases by 1.6 percentage points without CEI, compared to a 0 percentage point change in the baseline counterfactual exercise.

C.2 Labor Supply and Equilibrium

The labor supply side is modeled with a standard structure of occupation choice. Let L denote the number of ex-ante homogeneous workers. Workers observe the wage of each occupation determined in the market, w_o , occupation-specific utility ξ_o , and idiosyncratic utility realized for each occupation ζ . The worker chooses an occupation that gives the highest utility. All workers receive the same wage and utility for any given occupation. Consequently, once they choose an occupation, they are indifferent across industries. The occupation choice problem can be written as follows:

$$o^* = \operatorname{argmax}_o \{ \log w_o + \log \xi_o + \zeta \}.$$

Assuming that ζ follows an i.i.d. Type 1 Extreme Value Distribution with scale parameter $1/\eta$, the following equation determines the supply of occupational labor.

$$\frac{L_o}{L} = \frac{\exp(\eta \log w_o + \eta \xi_o)}{\sum_{o'} \exp(\eta \log w_{o'} + \eta \xi_{o'})}. \quad (\text{A18})$$

C.3 Elasticity of Substitution

Combining the first order conditions (7)–(9),

$$\begin{aligned}\frac{k_{dio}}{l_{io}} &= z_{dio} \left(\frac{r_{dio}}{w_o} \right)^{-\rho_d} \\ \frac{k_{sio}}{l_{io}} &= \tilde{\Theta}_{io}^{\frac{\rho_d - \rho_s}{\rho_d}} z_{sio} \left(\frac{r_{sio}}{w_o} \right)^{-\rho_s}, \text{ where} \\ \tilde{\Theta}_{io} &= \left(z_{dio} \left(\frac{r_{dio}}{w_o} \right)^{1-\rho_d} + 1 \right)^{\frac{\rho_d}{\rho_d - 1}}.\end{aligned}\tag{19}$$

Then,

$$\begin{aligned}d \log (k_{dio}/l_{io}) &= -\rho_d d \log (r_{dio}/w_o), \\ d \log (k_{sio}/l_{io}) &= -\rho_s d \log (r_{sio}/w_o) \\ &\quad + \frac{\rho_d - \rho_s}{\rho_d} \frac{d \log \Theta_{io}}{d \log (r_{dio}/w_o)} d \log (r_{dio}/w_o), \\ d \log \Theta_{io} &= -\rho_d \frac{k_{dio} r_{dio}}{k_{dio} r_{dio} + w_o l_{io}} d \log (r_{dio}/w_o).\end{aligned}\tag{20}$$

Combining these expressions gives Equation (15).

C.4 Equations for Counterfactual Equilibrium

The counterfactual exercise aims to derive the counterfactual equilibrium in the absence of CEI in 1980-2015. To isolate this effect, we fix patent counts ($\# \text{Patent}_{jion}$) at their observed 1980 levels while maintaining all other demand and supply components, including total employment (L), at their 2015 values.

Following [Caunedo et al. \(2023\)](#), we assume that a CES aggregator will make capital bundles with $\phi = 1.13$ as the substitution elasticity.

$$k_{jio} = \left(\sum_n \xi_{jion}^{\frac{1}{\phi}} x_{jion}^{\frac{\phi-1}{\phi}} \right)^{\frac{\phi}{\phi-1}}. \quad (\text{A21})$$

To solve for the counterfactual equilibrium, we utilize a system of equations that governs the allocation of labor across industries within an occupation. Specifically, we derive the following relationship from the first-order conditions of cost minimization with respect to labor inputs l_{io} .

$$1 = \frac{\alpha_i}{\alpha_j} \left(\frac{\mu_{io}}{\mu_{jo}} \right)^{\frac{1}{\sigma}} \left(\frac{Y_i}{Y_j} \right)^{\frac{1}{\sigma}-1} \left(\frac{y_{io}}{y_{jo}} \right)^{\frac{1}{\rho_s}-\frac{1}{\sigma}} \left(\frac{\tilde{\Theta}_{io}}{\tilde{\Theta}_{jo}} \right)^{\frac{\rho_s-\rho_d}{\rho_s\rho_d}} \left(\frac{l_{io}}{l_{jo}} \right)^{-\frac{1}{\rho_s}}, \quad (\text{A22})$$

$$Y_i = l_{io} \left(\sum_o \mu_{io}^{\frac{1}{\sigma}} \left(\frac{l_{io}}{l_{i0}} \right)^{\frac{\sigma-1}{\sigma}} \tilde{y}_{io}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} = l_{i0} \tilde{Y}_i. \quad (\text{A23})$$

Equation (A23) expresses industrial output as a linear function of l_{io} , the labor input of a reference occupation 0 and $\tilde{Y}_i$ which depends only on the ratio of labor inputs with respect to a reference occupation 0. The manager (OCC1990 = 22) is used as the reference occupation.

By combining Equations (A22) and (A23), we obtain the final equation used to determine industry-level employment for each occupation:

$$1 = \frac{\alpha_i}{\alpha_j} \left(\frac{\mu_{io}}{\mu_{jo}} \right)^{\frac{1}{\sigma}} \left(\frac{\tilde{Y}_i}{\tilde{Y}_j} \right)^{\frac{1}{\sigma}-1} \left(\frac{\tilde{y}_{io}}{\tilde{y}_{jo}} \right)^{\frac{1}{\rho_d}-\frac{1}{\sigma}} \left(\frac{\tilde{\Theta}_{io}}{\tilde{\Theta}_{jo}} \right)^{\frac{\rho_d-\rho_s}{(\rho_s-1)\rho_d}} \left(\frac{l_{io}}{l_{jo}} \right)^{-1}. \quad (\text{A24})$$

D Robustness

D.1 Alternative Thresholds and Citation Measures

This section evaluates the sensitivity of our structural estimates to the specific criteria used to distinguish between task-similar and task-dissimilar capital. Our baseline framework employs a 90th percentile threshold within the occupation-capital similarity score distribution to define task-similar capital, a choice that yields statistically significant differences in the CEI-s and CEI-d measures during structural regression.

Table A5 provides a comparison of these results against alternative thresholds and citation-based innovation measures. While the estimated coefficients for the CEI measures remain robust across various specifications, we find that the estimates of elasticities are more sensitive to the chosen threshold. Specifically, as the threshold is raised—thereby expanding the classification of task-dissimilar capital—the estimated elasticities of substitution between labor and capital tend to increase. Furthermore, utilizing citation-based measures of CEI rather than raw patent counts also results in higher estimated elasticities, suggesting that the qualitative weight of innovation may amplify the substitution effects observed in the baseline model.

Table A5: Estimation Results with Different Thresholds and Citations

	ρ_s	ρ_d	σ	γ_p	γ_s	γ_d
90th Percentile	1.63 (0.44)	1.24 (0.20)	1.59 (0.36)	-0.72 (0.02)	-0.74 (0.09)	1.57 (0.18)
89th Percentile	1.34 (0.13)	0.52 (0.13)	2.12 (0.42)	-0.72 (0.02)	-0.98 (0.12)	1.67 (0.24)
91st Percentile	2.05 (0.26)	2.02 (0.54)	2.84 (0.41)	-0.66 (0.02)	-0.80 (0.10)	1.88 (0.21)
Citation	1.92 (0.56)	1.60 (0.54)	1.72 (0.33)	-0.72 (0.02)	-0.78 (0.09)	1.44 (0.15)

Notes. This table shows estimates of structural parameters in Table 4 with alternative CEI measures and publication instruments. ρ_s (ρ_d) is the elasticity of substitution between task-similar (task-dissimilar) capital and labor. σ is the elasticity of substitution between different occupational services. γ_p is the coefficient of CEI on user costs of capital. γ_s (γ_d) is the coefficient of CEI-s (-d) on occupational service demand shifter. The rows labeled the 90th, 89th, and 91st Percentiles mean the percentiles of task-capital similarity scores used to calculate thresholds for similar-dissimilar distinction. The baseline uses the 90th percentile. The row labeled Citation uses the number of citations on the patents to measure CEI.

D.2 Reweighting Using Wikipedia Missing Rates

Because certain capital tools are not cataloged on Wikipedia, they are omitted from our primary text-matching analysis. This property introduces a potential source of measurement error; specifically, our measure of CEI could be systemically underestimated for occupations that rely heavily on capital goods with higher missing rates. To mitigate this potential selection bias, we perform a robustness check in which we reweight individual tools by the inverse of their respective Wikipedia match rates within each NIPA category.

Table A6 presents the structural parameter estimates obtained under this alternative reweighting scheme. The resulting estimates remain quantitatively and qualitatively close to our baseline findings.

Table A6: Parameter Estimates Under Reweighting

	ρ_s	ρ_d	σ	γ_p	γ_s	γ_d
Estimate	1.40	1.12	1.33	-0.67	-0.68	1.38
SE	(0.45)	(0.15)	(0.34)	(0.02)	(0.09)	(0.16)

Notes. This table shows the estimates and standard errors of the regression equations (17) and (18) when the tools are weighted by the inverse of the finding rate in Wikipedia. ρ_s (ρ_d) is the elasticity of substitution between task-similar (task-dissimilar) capital and labor. σ is the elasticity of substitution between different occupational services. γ_s (γ_d) is the coefficient of CEI-s (-d) on the demand shifter for occupational services.

E Counterfactuals

E.1 CEI and Task-Biased Labor Market Changes

We test what task-biased changes in the labor market would look like without CEI between 1980 and 2015 and thereby see if CEI constitutes task-biased technical changes in [Autor et al. \(2003\)](#). The task bias of labor market changes is measured in the following auxiliary regression that relates the log differences in wage and employment between 1980 and 2015 to abstract and routine task scores at the occupation level from [Autor and Dorn \(2013\)](#). The abstract and routine task scores with values between zero and one measure the intensity of abstract and routine tasks for each occupation.

$$\text{Change}_o = \theta_0 + \theta_1 \text{Task Score}_o + \varepsilon_o. \quad (\text{A25})$$

Each occupation is weighted by its employment in 1980. The OLS estimates for θ_1 are used as a measure of the task bias of changes in the labor market in 1980-2015, since they quantify the cross-sectional correlation between task scores with changes in wage and employment at the occupation level. Table [A7](#) shows the estimates for the regression coefficients. During this period, if an occupation has a one-standard-deviation higher score on abstract tasks, it has 0.17 and 0.44 standard deviations higher employment and wage growth rates, respectively.

In Panel A of Table [A7](#), two counterfactual equilibria are computed for $\eta = 0.3$ and $\eta = 1.5$ when the patent measures return to their levels in 1980. The biases in employment and wage are substantially smaller for both abstract and routine task scores in the absence of CEI. The effect on the wage is greater than the effect on employment, particularly when $\eta = 0.3$. This is due to the calibrated value of the supply elasticity η being too low to generate large responses in employment. Between 1980 and 2015, employment changes were significantly more dispersed than wage changes. Thus, with the values of supply elasticity considered, most employment changes result from non-wage supply shifters, ε_o . With $\eta = 1.5$, the employment responses become larger, while the wage responses decrease.

Table A7: Counterfactual: Task-Biased Labor Market Changes

	Abstract		Routine	
	Employment	Wage	Employment	Wage
Actual Change	0.17	0.44	-0.26	-0.33
<u>Panel A. Varying Supply Elasticity</u>				
Elasticity=0.3	0.08	-0.12	-0.21	0.02
Elasticity=1.5	-0.06	-0.05	-0.09	-0.03
<u>Panel B. Similar vs. Dissimilar CEI</u>				
Similar	0.07	0.13	-0.18	-0.16
Dissimilar	-0.00	0.02	-0.13	-0.06
<u>Panel C. Different Channels</u>				
Capital User Cost	0.18	0.44	-0.26	-0.33
Demand Shifter	-0.06	-0.05	-0.08	-0.02

Notes. This table shows the actual and counterfactual regression coefficients of the wage and employment growth rates at the occupation level in the occupational task scores of Autor and Dorn (2013) as in Equation (A25). The counterfactual equilibrium fixes the CEI measures at their levels in 1980. Panel C fixes patent measures to the 1980 level only when calculating changes in user costs and demand shifters, respectively. Panels B and C assume supply elasticity=1.5.

Panel B shows the counterfactual results, where only one type of CEI is fixed to the level in 1980. CEI-s and CEI-d have qualitatively similar effects on the task bias of employment and wage changes. CEI-s is higher for abstract and non-routine occupations, whereas CEI-d is higher for abstract ones. Consequently, the negative effect of CEI-s yields results analogous to the positive effect of CEI-d on occupational service demand. Given the larger magnitude of the coefficient estimate for γ_d in Table 4, CEI-d has larger quantitative effects.

Panel C decomposes the two channels in which CEI affects occupational labor demand. Almost all employment responses operate through demand shifters. Lower user costs of capital from CEI have limited effects on employment. This is because the estimated gaps between ρ_s , ρ_d , and σ are small. In Equation (10), the exponents of $\tilde{\Theta}_{io}$ and $\tilde{y}_{io}$ are $(\rho_s - \rho_d)/(\rho_s \times \rho_d) = 0.19$ and $-1/\sigma + 1/\rho_s = -0.02$, respectively. Together, a 1% increase in $\tilde{\Theta}_{io}$ increases the occupational labor demand in Equation (10) by less than 0.2%, and a 1% increase in $\tilde{y}_{io}$ reduces it only by 0.02%. Moreover, ρ_d and ρ_s are close to one,

weakening the effect of CEI-s and CEI-d on $\tilde{\Theta}_{io}$ and $\tilde{y}_{io}$ in Equations (11) and (12). Thus, the user cost channel has a minor role in generating significant employment responses. In contrast, the estimated coefficients of CEI on the demand shifter are large.

E.2 CEI and Labor Market Polarization

Table A8: Counterfactual Polarization

	Employment			Wage		
	Low	Middle	High	Low	Middle	High
Actual Change	0.17	-0.58	0.41	-0.03	-0.55	0.58
Panel A. Varying Supply Elasticity						
Elasticity = 0.3	0.25	-0.57	0.32	0.27	-0.25	-0.02
Elasticity = 1.5	0.32	-0.46	0.14	0.25	-0.31	0.06
Panel B. Similar vs. Dissimilar CEI						
CEI-s	0.34	-0.63	0.29	0.24	-0.52	0.28
CEI-d	0.30	-0.46	0.16	0.17	-0.23	0.06
Panel C. Different Channels						
Capital User Cost	0.16	-0.56	0.40	-0.03	-0.55	0.58
Demand Shifter	0.30	-0.41	0.11	0.24	-0.27	0.03

Notes. This table shows the actual and counterfactual growth rates of wage and employment when occupations are grouped by their wages in 1980. The counterfactual equilibrium fixes the CEI measures at their levels in 1980. The wage and employment changes are standardized to have a zero mean and a standard deviation of one in each case. Columns labeled Low and High denote occupations with 1980 wage levels in the first and fourth quintiles, respectively. Panel C fixes patent measures to the 1980 level only when calculating changes in user costs and demand shifters, respectively. Panels B and C assume supply elasticity=1.5.

Lastly, we study how CEI is related to polarization in the labor market after the 1980s. The first row of Table A8 summarizes wage and employment changes between 1980 and 2015 for three occupation groups by their residual wages in 1980. As in Autor and Dorn (2013), employment and wage changes at the occupation level take a U-shaped form over the wage level in 1980. High-wage occupations in the fifth quintile have 0.41 and 0.58 standard deviations higher employment and wage growth rates than the average occupations, respectively, and low-wage occupations also exhibit higher cross-sectional wage and employment growth rates than the average. The middle-wage ones, on the contrary, experience lower growth in terms of both employment and wage; the employment growth rate is lower by 0.58 standard deviations than the average, and the wage growth rate is lower by 0.55 standard deviations than the average.

Panel A of Table A8 shows the counterfactual employment and wage growth when

the patent measure, Patent_{jont} , is fixed at the level in 1980. Under the counterfactual equilibrium without CEI, wage and employment increase less for high-wage occupations. This result comes from high-wage occupations with higher CEI-d and lower CEI-s than others. The reduction in high-wage employment is driven more by increases in low-wage occupations than by middle-wage ones. CEI has a smaller impact on demand for middle-wage occupations because low-wage occupations have a higher level of CEI-s and a lower level of CEI-d than middle-wage occupations.

Panel B of Table A8 shows the polarization measures from counterfactual equilibria when only one type of capital has the patent measure fixed to 1980. Both CEI-s and CEI-d contribute to the growth of high-wage occupations, and the effects are larger for CEI-d because CEI-d is higher than CEI-s and the coefficient of CEI-d on demand shifter is larger.

The last panel summarizes the counterfactual results when the impact of CEI is active for only the user costs of capital or the demand shifter in the production function at each time. Similarly to the task bias exercise, the vast majority of wage and employment changes arise from the demand-shifter channel. This is also because the estimated gaps between elasticities of substitution are estimated to be too small to have substantial employment effects.

E.3 CEI and User Costs of Capital

In this section, we explore in more detail the relationship between CEI and user costs of capital and the role of user cost changes in labor market trends. First, we summarize changes in user costs associated with CEI. Then, we quantify how user cost changes alone contribute to trends in labor share for different occupations using the structural model estimated in Section 4.

Table A9 summarizes the actual and counterfactual user costs of capital for selected types of NIPA assets in 1980, 1990, 2000, and 2015. The CEI variables are redefined using patents granted until 1990, 2000, and 2015 relative to patents granted until 1980. Computers experience the sharpest decline in user costs over time. The baseline estimate $\gamma_p = -0.72$ is used to calculate counterfactual user costs. In addition to communication and photocopy equipment, the user costs of computers are the most heavily affected by CEI. Office equipment, metal products, and automobiles have the lowest CEI measures, and thus their user costs are the least affected by the absence of CEI.

Table A9: CEI and User Costs of Capital

	Actual				Without CEI			
	1980	1990	2000	2015	1980	1990	2000	2015
Computers	0.37	0.25	0.06	0.04	0.37	0.68	0.57	0.40
Communication eq.	0.14	0.15	0.11	0.06	0.14	0.28	0.48	0.40
Medical eq.	0.20	0.25	0.26	0.44	0.20	0.46	0.79	1.50
Photocopy eq.	0.24	0.26	0.20	0.32	0.24	0.72	1.32	3.29
Office eq.	0.40	0.46	0.47	0.66	0.40	0.58	0.89	1.22
Metal products	0.16	0.17	0.21	0.65	0.16	0.27	0.42	1.23
Engines	0.14	0.14	0.20	0.58	0.14	0.21	0.52	2.15
Aircraft	0.10	0.15	0.21	0.87	0.10	0.25	0.59	4.05
Electrical eq.	0.23	0.28	0.31	0.60	0.23	0.48	0.88	2.42
Autos	0.23	0.32	0.37	0.70	0.23	0.45	0.77	1.59
Ag. Machinery	0.17	0.22	0.27	0.78	0.17	0.30	0.54	1.73
Serv. Machinery	0.20	0.26	0.37	0.70	0.23	0.45	0.77	1.59

Notes. This table shows the average and counterfactual user costs of capital for selected categories of NIPA capital over time. User costs are rental costs of capital goods equivalent to one dollar in 1980. Counterfactual average user costs are calculated using patent measures from 1980 relative to 2015 with a price elasticity of $\gamma_p = -0.72$. ‘eq.’ means equipment.

We repeat the counterfactual exercise with the level of capital user costs fixed at their

levels in 1980. Table A10 displays the changes in the aggregate labor share and their decomposition in the data and in counterfactual equilibria without changes in user costs. User costs also do not reduce, but increase the aggregate labor share.

Table A10: Declines in Labor Share without User Cost Changes

	Within	Between	All
Actual Change	0.00	-2.68	-2.68
Elasticity=0.3	-3.23	-0.49	-3.71
Elasticity=1.5	-3.22	-0.49	-3.71

Notes. This table shows the actual and counterfactual changes in the labor share between 1980 and 2015. The counterfactual equilibrium fixes the user costs of all capital inputs at their levels in 1980. Changes in labor share within and between occupations are derived in Equation (22). Two rows have different values for elasticity of occupational labor supply.

The largest difference between CEI and user costs in decomposing the aggregate labor share comes from within-occupation labor shares. In Figure A9, CEI reduces the labor shares of occupations in the first and second quintiles of relative labor costs, but increases the labor shares in other groups. Heterogeneous effects on user costs of capital reallocate factor income from occupations with low labor shares to those with high labor shares, as CEI does in Figure 4.

Occupations with high wages relative to capital costs experience increases in occupational labor shares because declines in user costs are heterogeneous. In Figure A10, user costs of task-dissimilar capital decreased more for occupations with low relative labor costs. User costs of task-similar capital decreased relatively more uniformly across relative labor costs. Consequently, lower user costs compress the relative demand for services produced by high-labor-cost occupations. Because the supply of labor is less elastic than that of capital, wages reduce more than user costs of capital for these occupations, which drives up their within-occupation labor shares. Since these occupations tend to have a lower labor share, their reduced factor income share contributed to a higher aggregate labor share in the between-occupation margin.

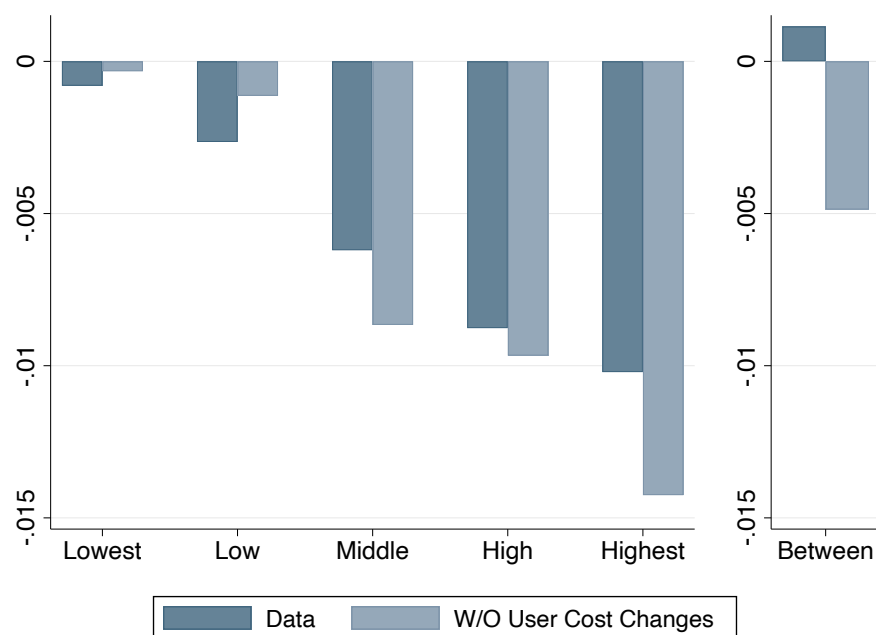


Figure A9: Role of Declining User Costs in Within-Occupation Labor Share Change

Notes. This graph shows the average within-occupation labor shares of equal-sized groups of occupations by their relative labor cost in 1980 in data and counterfactual equilibrium with user costs of capital fixed in 1980. Occupations are weighted with their factor income share in 1980.

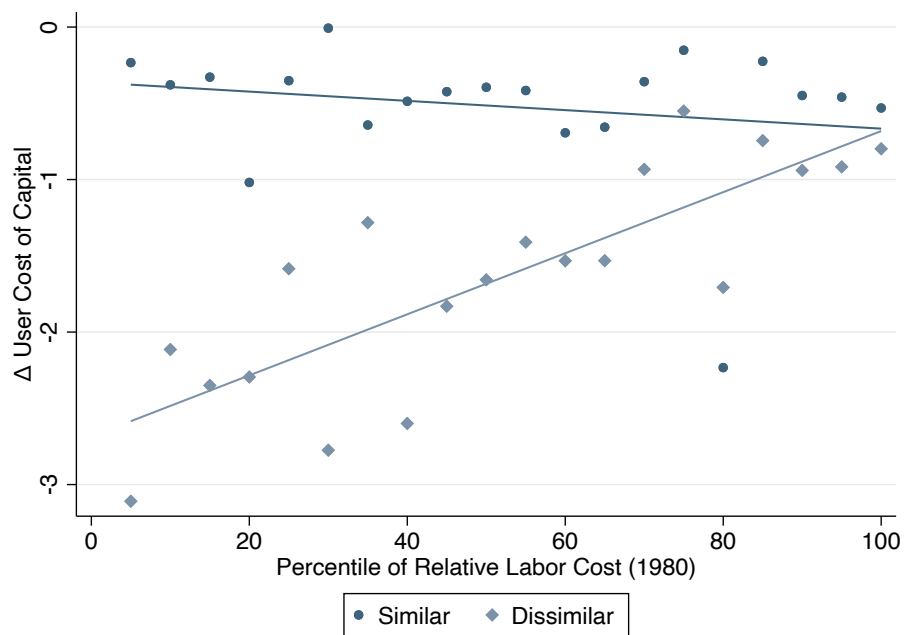


Figure A10: User Cost Changes

Notes. This graph shows the average changes in user costs of task-similar and task-dissimilar capital by their relative labor cost in 1980. Occupations are weighted with their factor income share in 1980.

E.4 Computers and Robots

This section quantifies the role of computer- and robot-related technologies within our broader CEI framework, comparing our findings with seminal studies that treat these two specific categories as the primary drivers of modern labor market restructuring (Autor et al., 2003; Acemoglu and Restrepo, 2018b; Burstein et al., 2019). This section first summarizes the importance of computers and robots in CEI measures. Then, we repeat the counterfactual exercise after setting only innovations related to computers fixed at the level in 1980. Since robots account for a negligible fraction of CEI, we do not repeat the counterfactual exercise that sets only innovations related to robots fixed at the level in 1980.

Table A11: Share of Computer and Robot in Matched Patents (%)

	1970	1980	1990	2000	2010
Computer	1.56	3.05	9.86	10.13	8.92
Robot	0.27	0.31	0.22	0.22	0.21

Notes. This table displays the shares of computers and robots in patents matched to capital goods in the UNSPSC. The column titles indicate the decades for which the patents were applied, except under 2010. Column 2010 collects patents applied from 2011 to 2015.

To operationalize this analysis, we identify computer capital based on the “computer equipment and accessories” product family within the UNSPSC classification system (43210000), while robotics and automation assets are identified by selecting equipment descriptors containing the terms “automatic,” “robot,” or “drone”. As documented in Table A11, patents matched to computers accounted for only 1.6% of matched patenting activity in the 1970s, but grew steadily over time, ultimately constituting roughly 10% of all utility patent applications filed after the 1990s. In sharp contrast, robots represent a small fraction of our aggregate measure, consistently accounting for just 0.2% to 0.3% of matched patents across all historical intervals. Due to this scarcity of explicitly robot-related patenting data, we omit robots from the subsequent counterfactual exercises and focus our analysis entirely on computers.

Table A12 summarizes the distribution of computer expenditure intensity across occu-

pation groups, demonstrating that computers function primarily as task-dissimilar capital for the vast majority of occupations. Over our sample period, the share of computers in total capital increased significantly, reaching 37% of aggregate equipment expenditures by 2015, with the fastest growth concentrated among high-wage, high-labor-share occupations intensive in abstract and non-routine tasks.

Table A12: Computer Intensity across Occupation Groups

	Similar			Dissimilar		
	Low	Middle	High	Low	Middle	High
<u>Panel A. Across Abstract Score</u>						
1980	0.08	4.79	7.52	13.45	23.81	29.58
2015	0.06	5.11	5.88	13.45	27.29	31.07
<u>Panel B. Across Routine Score</u>						
1980	0.04	6.77	1.49	31.09	19.01	24.88
2015	0.04	6.45	1.85	43.57	16.85	30.00
<u>Panel C. Across Labor Share in 1980</u>						
1980	0.60	6.64	1.95	3.02	28.32	29.48
2015	0.19	5.48	5.17	1.02	28.43	42.84
<u>Panel D. Across Wage in 1980</u>						
1980	0.00	5.02	6.71	20.94	21.89	27.08
2015	0.00	5.44	4.86	18.66	26.14	28.70

Notes. This table summarizes the share of computers in task-similar and task-dissimilar capital expenditures over occupations grouped by task scores, labor shares, and wage levels. The columns labeled Low and High represent the occupations in the first and fifth quintiles, respectively, while those labeled Middle cover occupations within the second to fourth quintiles. Numbers in percent.

This expansion is mirrored in the CEI measures reported in Table A13. Computers accounted for a small fraction of patents on task-dissimilar capital for all occupational groups in 1980. In 2015, computers represented a large fraction of patents related to task-dissimilar capital, except for occupations with low abstract scores, low labor shares, and low wages. Outside of these groups, the importance of computer-related patents is close to uniform across occupations.

To quantify the impact of computer-related innovation on macroeconomic changes, we re-calculate the counterfactual equilibrium without changes in computer-related patents

Table A13: Share of Computer-Based Patents across Occupation Groups

	Similar			Dissimilar		
	Low	Middle	High	Low	Middle	High
<u>Panel A. Across Abstract Score</u>						
1980	0.00	5.52	1.84	0.95	7.53	18.93
2015	0.00	7.23	2.98	3.74	18.07	25.81
<u>Panel B. Across Routine Score</u>						
1980	0.00	4.28	1.14	12.76	8.61	2.33
2015	0.00	5.42	2.95	26.31	13.11	16.14
<u>Panel C. Across Labor Share in 1980</u>						
1980	0.03	5.15	3.81	0.53	8.75	15.88
2015	0.01	6.45	6.60	0.92	18.65	27.90
<u>Panel D. Across Wage in 1980</u>						
1980	0.00	4.95	2.40	3.08	7.77	15.33
2015	0.00	6.72	2.82	6.29	17.01	25.71

Notes. This table summarizes the share of computers in matched patents of occupations grouped by task scores, labor shares, and wage levels. The columns labeled Low and High represent the occupations in the first and fifth quintiles, respectively, while those labeled Middle cover occupations within the second to fourth quintiles. The numbers are in percent.

between 1980 and 2015. All other patents remain in 2015. Table A14 shows the changes in the labor shares without computer-based CEI. Without increases in computer-related patents over time, the declines in labor share would have been smaller, but with a much smaller magnitude than in Table 5.

Table A14: Declines in Labor Share without Computer-based CEI

	Within	Between	All
Actual Change	-2.79	0.11	-2.68
Elasticity = 0.3	-1.98	-0.62	-2.60
Elasticity = 1.5	-2.08	0.16	-1.93

Notes. This table shows the actual and counterfactual regression coefficients of the wage and employment growth rates at the occupation level in the occupational task scores of Autor and Dorn (2013) as in Equation (A25). The counterfactual equilibrium uses CEI measures that fix only patents related to computers at their levels in 1980. Two rows have different values for elasticity of occupational labor supply.

Table A15 displays the coefficient estimates of task scores on wage and employment

changes in counterfactual equilibrium. Similarly to the labor share exercise, the magnitude of the change is smaller than in Table A7. This is because computers account for less than 30% of capital expenditures and CEI, almost all occupations use computers, and computers are almost uniformly important in CEI measures.

Table A15: Task-Biased Labor Market Changes without Computer-based CEI

	Abstract		Routine	
	Employment	Wage	Employment	Wage
Actual Change	0.17	0.44	-0.26	-0.33
Elasticity = 0.3	0.14	-0.02	-0.23	-0.03
Elasticity = 1.5	0.04	0.08	-0.15	-0.10

Notes. This table shows the actual and counterfactual regression coefficients of the wage and employment growth rates at the occupation level in the occupational task scores of Autor and Dorn (2013) as in Equation (A25). The counterfactual equilibrium uses CEI measures that fix only patents unrelated to computers at their levels in 1980. Two rows have different values for elasticity of occupational labor supply.

Lastly, Table A16 describes counterfactual changes in employment and wage growth between occupations grouped by their wages in 1980. The results also imply that computers generated labor market polarization, but with a smaller magnitude than CEI.

Table A16: Polarization without Computer-based CEI

	Employment			Wage		
	Low	Middle	High	Low	Middle	High
Actual Change	0.20	-0.61	0.41	-0.04	-0.53	0.57
Elasticity = 0.3	0.24	-0.59	0.35	0.18	-0.22	0.04
Elasticity = 1.5	0.28	-0.49	0.21	0.15	-0.30	0.15

Notes. This table shows the actual and counterfactual growth rates of the wage and employment growth of occupations grouped by their wages in 1980. The counterfactual equilibrium uses CEI measures that fix only patents unrelated to computers at their levels in 1980. The wage and employment changes are subtracted from the mean and divided by the standard deviation of the changes at the occupation level in each case. Columns labeled Low and High denote occupations with 1980 wage levels in the first and fifth quintiles, respectively. The columns labeled middle denote occupations between the two quintiles. Two rows have different values for elasticity of occupational labor supply.

E.5 Aggregate ES between Labor and Capital

We follow the steps in [Oberfield and Raval \(2021\)](#) to define the aggregate substitution elasticity between labor and capital. Since our model assumes an elasticity of substitution within the industry of σ and an elasticity of substitution between industries of one, we take two steps to aggregate the substitution elasticity. The aggregation takes the elasticity of substitution at the occupation-level, $\hat{\rho}_{iot}$ in Equation (15).

$$\eta_{it} = (1 - \chi_{it}) \bar{\rho}_{it} + \sigma \chi_{it}, \quad (26)$$

$$\chi_{it} = \sum_o \frac{(\alpha_{iot} - \alpha_{it})^2 \theta_{iot}}{\alpha_{it} (1 - \alpha_{it})},$$

$$\bar{\rho}_{it} = \sum_o \frac{\alpha_{iot} (1 - \alpha_{iot}) \theta_{iot}}{\sum_{o'} \alpha_{io't} (1 - \alpha_{io't}) \theta_{io't}} \hat{\rho}_{io't}.$$

In this equation, $\alpha_{iot} = (r_{sio't}k_{sio't} + r_{dio't}k_{dio't}) / (w_{ot}l_{iot} + r_{sio't}k_{sio't} + r_{dio't}k_{dio't})$ is the capital share of factor income for occupation o within industry i , and $\theta_{iot} = (w_{ot}l_{iot} + r_{sio't}k_{sio't} + r_{dio't}k_{dio't}) / \sum_{o'} (w_{o't}l_{io't} + r_{sio't}k_{sio't} + r_{dio't}k_{dio't})$ is the factor income share of occupation o in industry i . Lastly, $\alpha_{it} = \sum_i \alpha_{iot} \theta_{iot}$ is the capital share of the factor income at the industry level.

Aggregation between industries gives

$$\eta_t^A = (1 - \chi_t) \bar{\eta}_t + \chi_t, \quad (A27)$$

$$\chi_t = \sum_i \frac{(\alpha_{it} - \alpha_t)^2 \theta_{it}}{\alpha_t (1 - \alpha_t)}, \quad (A28)$$

$$\bar{\eta}_t = \sum_i \frac{\alpha_{it} (1 - \alpha_{it}) \theta_{it}}{\sum_{i'} \alpha_{i't} (1 - \alpha_{i't}) \theta_{i't}} \bar{\eta}_{it}. \quad (A29)$$

$\theta_{it} = \sum_o (w_{ot}l_{iot} + r_{sio't}k_{sio't} + r_{dio't}k_{dio't}) / \sum_{i'o'} (w_{o't}l_{io't} + r_{sio't}k_{sio't} + r_{dio't}k_{dio't})$ is the factor income share of industry i , and α_t is the aggregate capital share of factor income.

The weight of the inter-industry elasticity of substitution χ_t is called the heterogeneity index in [Oberfield and Raval \(2021\)](#) because the index increases with the dispersion of capital expenditure share across industries and is zero if all industries have the same



Figure A11: Aggregate ES between Capital and Labor

capital expenditure share. Similarly, χ_{it} increases if occupations within an industry are heterogeneous in terms of capital expenditure share.

Figure A11 shows the aggregate elasticity of substitution between labor and capital. The aggregate ES decreased before 1990 mainly because the share of task-similar capital decreased, as shown in Figure A12. After 1990, on the other hand, the increase in the heterogeneity index contributed to a reduction in aggregate elasticity (See Panel A of Figure A13). The information and computer-manufacturing industries increased capital disproportionately more than the labor force, while the professional services industries reduced capital intensity. The increased heterogeneity of capital intensity between industries raised aggregate ES since the demand elasticity between industries is lower than the elasticity of substitution between capital and labor within each industry.

Figure A11 shows the aggregate elasticity in the reduced-form of substitution between labor and capital. Similarly to Oberfield and Raval (2021), the aggregate elasticity tends to decrease over time. The aggregate ES decreased in the 1980s mainly because the share of task-similar capital decreased, as in Figure A12. Changes in heterogeneity indices contributed little.

After the 1980s, however, the increase in the heterogeneity index led to a reduction in the aggregate elasticity (See Panel A of Figure A13). Information services and computer-manufacturing industries increased capital expenditures disproportionately more than

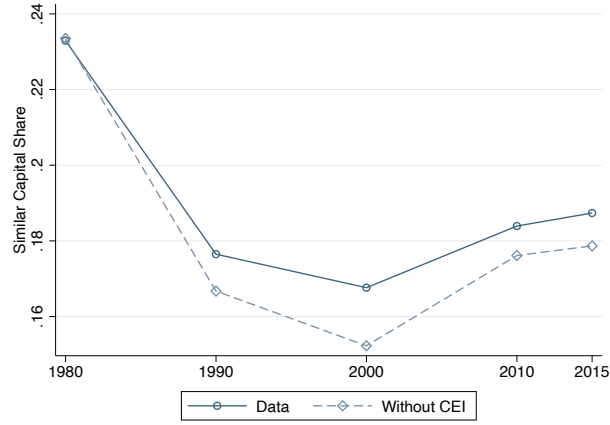


Figure A12: Similar Capital Share

the wage bill, whereas the professional services industries reduced capital share. The increased heterogeneity of capital shares between industries reduced aggregate ES since demand elasticity between industries is lower than the ES between capital and labor within each industry. On the other hand, because the ES between occupations within an industry, $\sigma = 1.6$, tends to be greater than the ES between capital and labor within an occupation, $\hat{\rho}_{io}$, the increase in χ_{it} after 1990 in Panel B of Figure A13 contributed to an increase in the aggregate ES.

Figure A13: Heterogeneity Index Between and Within Industry

